FEDERAL PURCHASE DESCRIPTION

BUS-TRANSIT-HEAVY DUTY-LOW FLOOR-40 FT.-34 PASSENGER –DIESEL ELECTRIC HYBRID

1.0 Scope and Classification

1.1 Scope

This specification defines requirements for heavy duty, 40 foot-long, minimum 34 passengers, perimeter seating configuration, low floor, diesel-electric hybrid buses. The bus shall be low floor design, be a maximum of 102 inches wide, have a left-hand operator location, and be fully ADA-compliant.

In addition, the bus shall have a minimum expected life of 12 years/500,000 miles and be intended for the widest possible spectrum of passengers, including children, adults, the elderly, and persons with disabilities. The bus shall meet other design goals, including long cycle life, low life-cycle cost, safety, maintainability, durability, and robust diagnostics.

This Purchase Description does not include all varieties of the commodity indicated by the title, but is intended to cover those vehicles generally acquired by the Government. Buses with standardized components and equipment are highlighted in this Standard. A selection of coded optional additional systems and equipment are included for agencies' divergent geographic and operational related needs.

1.2 Classification

The individual bus configuration classifications are referred to by a Standard Item Number (SIN) The SINs covered by this specification are as follows:

377C-Bus-Transit-Heavy Duty-Low Floor-40 Ft.-34 Passenger-Diesel Electric Hybrid

2.0 Applicable Documents

The vehicles shall conform to the latest revision of the following:

<u>Federal Regulations and Standards</u>:

Federal Motor Carrier Safety Regulations (FMCSR), 49CFR 393
Federal Motor Vehicle Safety Standards (FMVSS), 49CFR 571
Americans with Disabilities Act, 49 CFR 38
FTA Docket 90-A, October 20, 1993

Application for copies of DOT publications should reference the code of Federal Regulations, 49 CFR, and the Federal Register and should be addressed to the Superintendent of

Documents, U.S. Government Printing Office, Washington, DC 20402. They may also be accessed on the Internet through GPO Access at http://www.access.gpo.gov.

The body of this Purchase Description (PD) contains requirements relative to the following standards and recommended practices, and the offeror's proposed vehicles shall conform to all such requirements.

State Regulations and Standards

California Air Resources Board (CARB) http://www.arb.ca.gov/homepage.htm

CARB requirements may be obtained from:

California Air Resources Board
1001 "I" Street
PO Box 2815
Sacramento, CA 95812
Phone: (916) 322-2990

Society of Automotive Engineers (SAE) Standards and Recommended Practices:

J10 Automotive and Off-Highway Air Brake Reservoir Performance and Identification Requirements
J287 Driver Hand Control Reach
J516 Hydraulic Hose Fittings
J517 Hydraulic Hose
J534 Lubrication fittings
J541 Voltage Drop for Starting Motor Circuits
J553 Circuit Breakers
J678 Speedometers and Tachometers—Automotive
J941 Motor Vehicle Drivers' Eye Locations
J1050 Describing and Measuring the Driver's Field of View
J1052 Motor Vehicle Driver and Passenger Head Position
J1131 Performance Requirements for SAE J844 Nonmetallic Tubing and Fitting Assemblies Used in Automotive Air Brake Systems
J1149 Metallic Air Brake Tubing and Pipe
J1516 Accommodation Tool Reference Point
J1522 Truck Driver Stomach Position
J1625 Heavy Duty Circuit Breakers
J1708 Serial Data Communications Between Microcomputer Systems in Heavy-Duty Vehicle Applications

J163 Low tension Wiring and Cable Terminals and Splice Clips
J198 Windshield Wiper Systems- Trucks, Buses, and Multi-Purpose Vehicles
J382 Windshield Defrosting Systems Performance Requirement-Trucks, Buses, and Multi-Purpose Vehicles
J593 Backup Lamps (Reversing Lamps)
J673 Automotive Safety Glasses
J844 Nonmetallic Air Brake System Tubing
J683 Tire Chain Clearance
J686 Motor Vehicle License Plates
J994 Alarm—Backup—Electric Laboratory Performance Testing
J1127 Battery Cable
J1128 Low Tension Primary Cable
J1273 Recommended Practices for Hydraulic Hose Assemblies
J1292 Automobile, Truck, Truck-Tractor-Trailer, and Motor Coach Wiring

Application for copies of SAE publications should be addressed to:
SAE World Headquarters http://www.sae.org/
400 Commonwealth Drive
Warrendale, PA 15096-0001

Industry Standards

National Fire Protection Association (NFPA) http://www.nfpa.org/

NFPA 70 (National Electrical Code-NEC)
NFPA Standard 17 (Dry Chemical Fire Suppression Systems)
NFPA 52 Vehicular Fuel Systems Code
NFPA standards may be obtained from:
National Fire Protection Association
1 Batterymarch Park
Quincy, MA 02169-7471

American Society for Testing and Materials (ASTM) http://www.astm.org/

D-6751 Standard Specification for Biodiesel Fuel Blend Stock (B100) for Middle Distillate Fuels

ASTM standards may be obtained from:

ASTM International
100 Barr Harbor Drive

West Conshohocken, PA 19428-2959
Phone 610-832-9585

American Public Transportation Association (APTA) http://www.apta.com/Pages/default.aspx

APTA recommendations may be obtained from:

APTA
1666 K Street, NW
Suite 1100
Washington, DC 20006
Telephone: (202) 496-4800

National Institute for Automotive Service Excellence (ASE) http://ase.com/

ASE requirements may be obtained from:

National Institute for Automotive Service Excellence
101 Blue Seal Drive, S.E.
Suite 101
Leesburg, VA 20175
Phone 703-669-6600

Underwriters Laboratory, Inc. (UL) http://www.ul.com/

UL Standard 1254 UL Standard for Safety for Pre-Engineered Dry Chemical Extinguishing System Units

Underwriters Laboratory standards may be obtained from:

2600 N.W. Lake Rd.
Camas, WA 98607-8542
Telephone: 1.877.UL.HELPS (1.877.854.3577) or
http://ulstandardsinfonet.ul.com/catalog/stdscatframe.html

3.0 Overall Requirements

The bus shall comply with all applicable federal, state, and local regulations. These shall include, but not be limited to, the Federal Americans with Disabilities Act (ADA), Federal Motor Vehicle

Safety Standards and Regulations 49CFR 571 (FMVSS), 49CFR 393 (FMCSR), National Fire Protection Association (NFPA), United States Environmental Protection Agency (USEPA), California Air Resources Board (CARB) regulations, as well as state and local accessibility, safety and security requirements.

The offeror shall certify to the Government that the vehicle platform complies with 49 CFR, Bus Testing Rule. The offeror shall test the bus platform proposed at the Altoona (PA) Testing Facility and provide copies of all testing results, forms, and documentation as part of the proposal offer. The bus vehicle platform proposed shall be tested in accordance with the Altoona testing category requirements for heavy-duty large buses (minimum service life of 12 years or 500,000 miles). If the bus platform proposed by the offeror has already been tested successfully at the Altoona Testing Facility, the offeror shall provide verification of the successful testing with the offer.

The offeror shall explicitly and completely identify any and all exceptions taken to the requirements of this specification. Drawings, literature, and any other information submitted with an offer shall not constitute a stated or implied exception unless specifically identified in writing as an exception and accepted and implemented by an amendment to the solicitation or modification to the contract. Any exception deemed acceptable by the Government shall only apply to the specific item, requirement, etc. cited, and shall not extend to any other requirement.

Materials shall be as specified herein. When materials are not specified, the vehicles and all parts thereof shall be furnished to provide the intended function, durability, safety, and maintain good long-term appearance. All materials shall be new, shall be suitable for the intended purpose, and shall be free of any characteristics or defects in material and workmanship, which may affect the performance, function, durability, and serviceability of the finished vehicle, or detract from its appearance. The Government reserves the right to make the final determination of the suitability of all components and their arrangement on and in the vehicle.

The offeror responding to this solicitation shall be a bus body and chassis OEM- Original Equipment Manufacturer (see Paragraph 4.2 Definitions), or an OEM authorized dealer. The offeror shall have demonstrated the commercial manufacture and sale of low floor diesel-electric hybrid transit buses of designs equal to or of greater complexity than the requirements herein. Commercial order and sales data shall be submitted with the offer.

The offeror is encouraged to submit with its offer value engineering information (refer to Federal Acquisition Regulation [FAR] Part 48) in areas of vehicle design, construction, space utilization, etc. which will improve the quality, function, durability, and performance of the vehicle and/or provide a cost savings to the Government. Adoption of these submissions will be determined by the Government, and will only be implemented by an amendment to the solicitation or modification to the contract.

The offeror shall submit with its offer drawings that show the following:

- Interior plan view of the proposed bus, including seating arrangement, wheelchair locations, and dimensions verifying compliance with the requirements herein.
- Exterior front, rear, and side view with overall dimensions.
- Estimated curb weight and maximum GVW with total seated and standee passengers plus driver as required in Section 4.2-Definitions for the gross load to utilize in determining minimum GVWR and GAWR. The total number of seated and standee passengers based on this definition of gross load shall be furnished with the offer.
- Estimated weight distribution at front and rear axles at maximum GVW.

The offeror shall ensure that the application and installation of major bus subcomponents and systems are compliant with subcomponent vendors' requirements and recommendations. Components used in the vehicles shall be of heavy-duty design and proven in transit service.

3.0.1 Dimensions

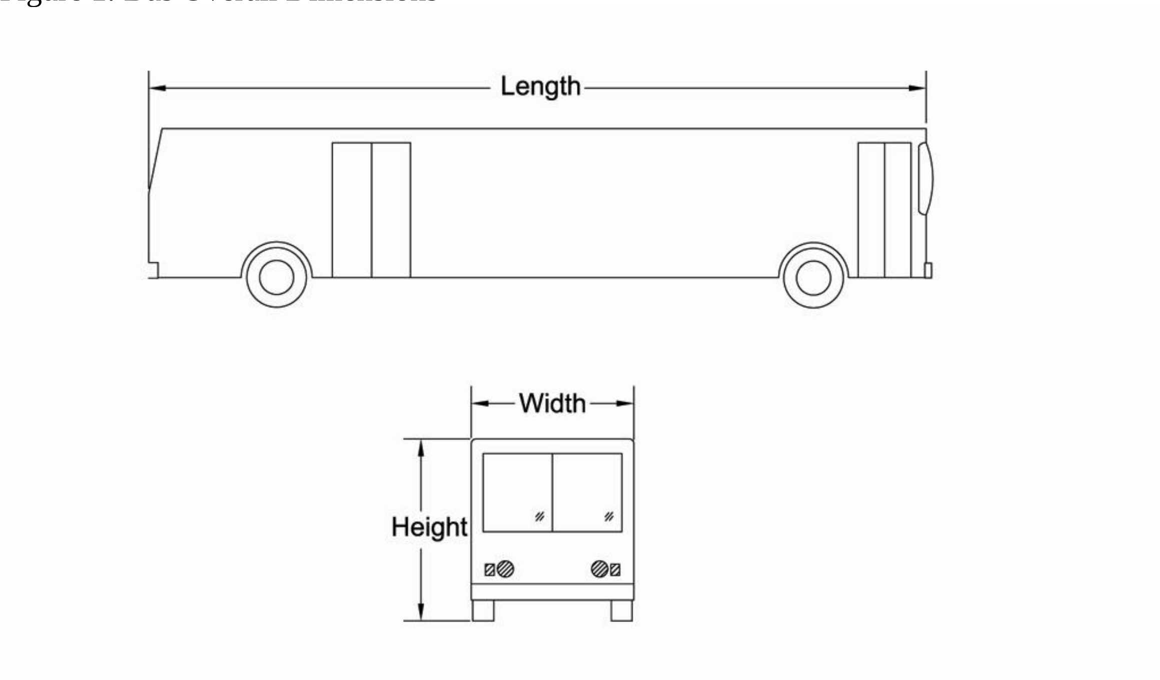
3.0.1.1 Physical Size

With the exception of exterior mirrors, marker and signal lights, bumpers, fender skirts, and rub rail, the buses shall have the following overall dimensions (as illustrated in Figure 1 below).

Bus Overall Dimensions:

• Length (excluding bumpers)	40 feet (± 6 inches)
• Width (exterior, excluding mirrors)	102 inches (+ 0/- 1 inches)
• Overall Height	135 inches (max.)

Figure 1: Bus Overall Dimensions



3.0.1.2 Underbody Clearance

The bus shall maintain the minimum clearances shown in Figure 2 regardless of load up to the gross vehicle weight rating. All components under the bus (including the oil pan) shall be protected from any impact with foreign objects.

- Wheel Area Clearance - Wheel area clearance, shall be 8 inches (minimum) for parts fixed to the bus body and 5 inches (minimum) for parts that move vertically with the axles.

- Ramp Clearances - Approach angle shall be 8 degrees (minimum). Breakover angle shall be 8 degrees (minimum). Departure angle shall be 8 degrees (minimum). Any encroachment into the approach or departure angle area shall encounter a structural member before any operating component or subsystem.

- Axle Zone Clearance - Axle zone clearance, which is the projected area between tires and wheels on the same axial centerline, shall be as great as possible within the confines of overall vehicle width.

- Ground Clearance - Ground clearance shall be 9 inches (minimum), except within the axle zone and wheel area.

3.0.1.3 Weight

3.0.1.3.1 Curb Weight

Curb weight of the bus shall be minimized to the greatest extent practicable while maximizing its integrity and durability.

Weight distribution shall be such that no axle shall be loaded past its GAWR at maximum gross vehicle weight with the gross load weight distribution in accordance with all applicable State and Federal requirements.

The Contractor shall submit to the Government during vehicle delivery, a certified weight slip for the curb weight of each axle and total vehicle weight.

3.0.1.3.2 GVWR and Capacity

The Gross Vehicle Weight Rating of the bus shall exceed the curb weight of the bus plus the capacity of passengers and driver. Total passenger capacity shall be the number of people (combination seated/standing) considering a minimum of 34 passenger seats that can be carried with the offeror's maximum rated (gross axle weight rating) front and rear axle, using 150 lbs./passenger and driver. See also Section 4.2-Definitions for the gross load. The GVW and axle loadings shall not exceed the GVWR and front and rear GAWR at maximum capacity. For purposes of calculations, the standee passengers may be considered to be equally distributed on the bus in the area allowed for standees.

The offeror shall furnish, with the offer, the maximum front and rear GAWR, the GVWR, and the maximum number of passengers (seated and standee) the bus can carry without exceeding the GAWR and GVWR at the 150 lb. /passenger and driver requirement.

3.0.1.3.3 Passenger Capacity Decal

An instruction decal shall be applied to the forward bulkhead visible to all passengers and driver. The lettering shall be not less than 1-1/2 inch in height and in strong contrast to the interior background color. The message can be in a single line or two lines and must state "Maximum Passenger Capacity XX" or "Capacity XX Passengers". The two X's shall represent the total seated and standee passenger capacity based on 3.0.1.3.2 above.

3.0.1.4 Life Expectancy

The bus shall be designed to operate in transit service for at least 12 years and/or 500,000 miles.

3.0.1.5 Maintainability Requirements

Routine, scheduled maintenance or inspection tasks as specified by the Contractor shall require an ASE (National Institute for Automotive Service Excellence) technician skill level of 3M or less (target). To facilitate ease of maintenance, the vehicle design shall maximize the use of line replaceable units (LRUs). All LRUs shall allow expedient replacement.
Readily accessible test ports shall be provided for routinely checked functions on-board the proposed vehicle (such as air intake, exhaust, hydraulic, pneumatic, charge-air, and engine cooling system(s)).

3.0.1.6 Material Selection/Interchangeability

The bus shall be designed using commercially available (off-the-shelf) components and materials to the greatest extent possible. For economy in maintenance, it is essential that all parts and units be arranged so that rapid assembly and disassembly will be possible for the vehicles being provided. The dimensions of all parts, unless particularly specified, shall be in accordance with current standards of the Society of Automotive Engineers (SAE), or the metric equivalent. All units or parts not specified shall be Contractor's standard units or parts and shall conform in material, design and workmanship to industry standards and shall meet or exceed all federal and state motor vehicle safety standards.

Components with identical functions shall be interchangeable to the extent possible. These components shall include passenger lamps (and components), window hardware, trim, interior components, seat assemblies and other components considered to be consumable. Components with non-identical functions shall not be, or appear to be, interchangeable. All parts shall be new and in no case will used, reconditioned, or obsolete parts be accepted. No advantages shall be taken by the manufacturer in the omission of any parts or details that make the vehicles complete and ready for service, even though such parts or details are not mentioned in these specifications.

3.0.1.7 Ambient Operation Requirements

The bus shall be engineered for operation in an ambient temperature range of -30 degrees to +115 degrees Fahrenheit (at relative humidity levels between 5 and 100 percent, and at altitudes up to 7,000 feet above sea level). Speed, gradeability, and acceleration performance requirements shall ideally be met at, or corrected to, 77 degrees Fahrenheit, 29.31 inches Hg, dry air per SAE J1995. Performance degradation is expected and allowed below -10 degrees Fahrenheit, ordering activities shall consider optional auxiliary heater (GSA option AH). Performance degradation is expected and allowed at elevations above 3000 feet above sea level. Ordering activities shall request, review, and consider OEM performance data prior to selection.

3.0.1.8 Elderly and Disabled Passenger Accessibility

The Contractor shall comply with all applicable Federal requirements defined in the Americans with Disabilities Act, 49 CFR Part 38, and all state and local regulations regarding mobility-impaired persons. Local regulations are defined as those below the state level.

3.0.2 Audible Noise Emission Control

Since noise levels are of a primary concern to the Government, the Government strongly prefers that each vehicle have a low level of exterior/interior noise and, as a design target, that each vehicle be significantly quieter than stated herein. If the offeror has optional sound deadening capability, the maximum sound deadening materials and insulation shall be furnished. The combination of inner and outer panels, plus any material used between them, and engine compartment insulation shall provide sufficient sound insulation so that sound levels do not exceed the values in sections 3.0.2.1 and 3.0.2.2 below.

3.0.2.1 Interior Noise Emissions

The interior noise shall be tested in accordance with Altoona Bus Testing interior noise test procedure 7.1.

A. With the left exterior of the bus exposed to a “white” noise minimum level of 83 dB(A), and the bus engine and all accessories “off”, the noise transmitted to the interior of the bus shall not exceed 75 dB(A).

B. Under full throttle acceleration from 0-35 MPH, and all accessories operating, the bus generated interior noise levels shall not exceed the values below:

- 85 dB(A) at any passenger seat location
- 78 dB(A) at the driver’s seat

3.0.2.2 Exterior Noise Emissions

The exterior noise shall be tested in accordance with Altoona Bus Testing exterior noise test procedure 7.2.

Airborne noise generated by the bus and measured from either side shall not exceed 83 dB(A) under full power acceleration when operated at or below 35 mph at curb weight and just prior to transmission upshift. The maximum noise level generated by the bus pulling away from a stop at full power shall not exceed 83 dB(A). The bus-generated noise, with bus stationary, from either side at low or high idle, with accessories and air conditioning “on” shall not exceed 65 dB(A). All noise readings shall be taken 50 feet from, and perpendicular to, the centerline of the bus. The pull away test shall begin with the front bumper even with the microphone. The stationary idle test shall be conducted with the rear bumper even with the microphone.

3.0.3 Back Up Alarm

The bus shall be furnished with an audible alarm that sounds whenever the ignition switch is in the "on" position and the drive is in reverse gear.

3.0.4 Electromagnetic Radiation (EMI)

Electrical and electronic sub-systems and components on all bus shall comply with FCC regulations.

Electrical and electronic sub-systems on the vehicles shall not be affected by external sources of RFI/EMI. This includes, but is not limited to, radio and TV transmission, portable electronic devices including computers in the vicinity of or onboard the bus, AC or DC power lines and RFI/EMI emissions from other vehicles.

Electronic components that are subject to RFI/EMI shall have shielded power and data cabling. Deviations from this requirement shall require Government approval. On-board equipment can include but is not limited to automatic passenger counters, two-way radio, electronic engine and transmission controls, automatic vehicle locating equipment, and destination signs and associated wiring.

If one vehicle and/or component is susceptible to RFI/EMI, it will be assumed that all vehicles suffer the same defect. Design corrections shall be made on a fleet basis. The Government reserves the right to review and approve proposed solutions.

Upon the request of the Government, the Contractor shall submit test data or other evidence that all of the above requirements are met.

3.0.5 Responsibility for Design

The contractor shall assume total and overall responsibility for the design and satisfactory operation of the vehicle and vehicle subsystems or component parts. The contractor's responsibility includes, but is not limited to, ensuring that the design and manufacture of the vehicle and its component parts are appropriate, coordinated, compatible, and perform correctly throughout the life of the vehicle (either together or individually).

3.0.6 OEM Requirements and Recommendations

The proposed bus vehicle is an assembly of components that may be sourced from several different original equipment manufacturers (OEMs) or suppliers. The Contractor shall obtain and adhere to the requirements and recommendations of the OEMs for the application and installation of components that are installed. In the absence of specific OEM requirements and/or recommendations, the Contractor shall obtain the OEM's approval for their installation. Components utilized in the vehicle shall be of heavy-duty design and proven in transit service.

When delivered the bus shall be in a ready-to-use condition. Lubricants, fuel, anti-freeze, and any other fluids utilized shall be of the type specified by the OEMs and installed at the levels recommended by the OEMs.

3.0.7 Technology Upgrades

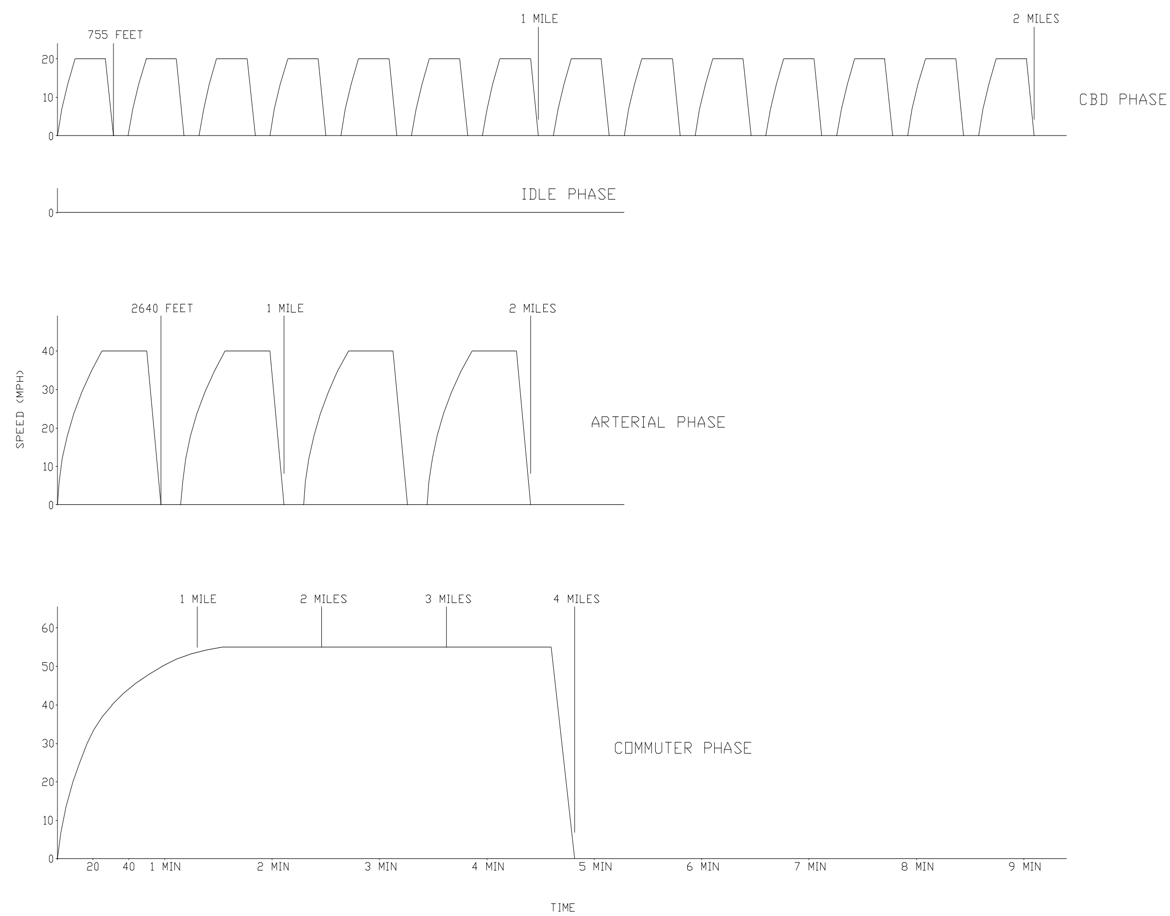
The offered bus vehicle shall blend service-proven design principles and equipment with the latest technological advancements without sacrificing reliability, maintainability, safety or performance.

The successful Contractor shall provide the Government with written notification (including a complete description of upgrade/advancement, trade-off analysis, and commercial implications) of viable technology upgrades and advancements throughout contract duration.

3.0.8 Operating Profile

The operating profile for design purposes shall consist of simulated transit type service. The duty cycle is described in the figure "Transit Bus Duty Cycle." The duty cycle consists of three phases to be repeated in sequence: a central business district (CBD) phase of 2 miles with 7 stops per mile and a top speed of 20 mph, an arterial route phase of 2 miles with 2 stops per mile and a top speed of 40 mph, and a commuter phase of 4 miles with 1 stop and a maximum speed of 55 mph and a 5 minute idle phase.

Phase	Stops/ Mile	Top Speed (mph)	Miles	Accel. Dist. (ft.)	Accel. Time (s)	Cruise Dist. (ft.)	Cruise Time (s)	Decel. Rate (fpsps)	Decel. Dist. (ft.)	Decel. Time (s)	Dwell Time (s)	Cycle Time (min-s)	Total Stops
CBD	7	20	2	155	10	540	18.5	6.78	60	4.5	7	9-20	14
Idle	-	-	-	-	-	-	-	-	-	-	-	5-0	-
Arterial	2	40	2	1035	29	1350	22.5	6.78	255	9	7	4-30	4
CBD	7	20	2	155	10	510	18.5	6.78	60	4.5	7	9-20	14
Arterial	2	40	2	1035	35	1350	22.5	6.78	255	9	7	4-30	4
CBD	7	20	2	155	10	510	18.5	6.78	60	4.5	7	9-20	14
Commuter	1 stop for phase	Max. or 55	4	5500	90	2 miles + 4580 ft.	188	6.78	480	12	20	5-10	1
Total			14									47-10	51
Average Speed - 17.8 mph													



Transit Bus Duty Cycle

3.1 Propulsion System and Chassis

3.1.1 Top Speed

The bus vehicle shall be capable of a top speed of 55 mph on a straight, level road at GVWR with all accessories operating. The maximum top speed shall be governed to not exceed 65 MPH.

3.1.2 Diesel Electric Hybrid Drive

The hybrid drive system shall a BAE Systems TB200 HybriDrive series hybrid drive system.

The hybrid drive system shall provide power to enable the proposed buses to meet the defined acceleration, top speed, gradeability, operate all propulsion driven accessories, while still maintaining passenger comfort and providing a smooth ride. Hybrid drive input torque rating shall exceed engine output torque. The hybrid drive shall be rated to operate at the GVWR of the bus.

The engine and hybrid drive combination shall incorporate electronically infinitely variable gear ratios that automatically optimize engine performance and economy during operation on all grades and operating conditions, and assure that all performance requirements herein are met. The hybrid drive and its push-button shift select control head shall be designed or interlocked so the possibility of damage or uncontrolled acceleration due to driver misuse of the shift select control head is minimized. The electronic control shall be able to utilize full electronic diagnostic techniques with an industrial grade testing instrument.

The electronically controlled hybrid drive shall have on-board diagnostic capabilities, be able to monitor functions, store and time stamp out-of-parameter conditions in memory, and communicate faults and vital conditions to service personnel. The hybrid drive shall contain built-in protection software to guard against severe damage. A diagnostic reader device connector port, suitably protected against dirt and moisture, shall be provided in the operator’s area.

An oil fill location accessible from the rear of the engine compartment shall be furnished. The drive unit shall also be furnished with a Titan Laboratories brass Probalyzer, or equal, fluid sampling plug installed in a location easily accessible for obtaining an oil sample.

Heaters and pumps requiring engine coolant shall have quarter-turn ball valves on each side of the device.

3.1.2.1 Top Speed

The bus vehicle shall be capable of a top speed of 55 mph on a straight, level road at GVWR with all accessories operating. The maximum top speed shall be propulsion system governed to not exceed 62 MPH.

3.1.2.2 Gradeability

Gradeability requirements shall be met on grades with a dry commercial asphalt or concrete pavement at GVWR with all accessories operating. The propulsion system shall enable the bus to maintain a speed of 44 mph on a 2-1/2 percent grade and 7 mph on a 16 percent grade. At a minimum, while in reverse, the buses shall be able to climb a 16 percent grade.

3.2.1.3 Acceleration

The acceleration rate shall meet the requirements below and be sufficiently gradual and smooth to prevent throwing standing passengers off-balance. Acceleration measurement shall commence when the accelerator is depressed. The minimum acceleration rates shall be as follows:

Speed (MPH)	Maximum Elapsed Time (Seconds)
10	5.0
20	10.1
30	19.0
40	31.0

3.1.2.4 Operating Range

The operating range of the proposed bus vehicle shall be a minimum of 360 miles with a full tank of diesel fuel, based on a combination of CBD, Arterial, and Commuter cycles, as developed by the FTA/Altoona.

3.1.2.5 Hybrid System Controller

The hybrid system controller shall monitor system inputs and execute outputs as appropriate to control the operation of all hybrid devices. This controller shall directly control power electronics necessary for operation of the engine, generators/electric motors, energy storage and other related hybrid devices.

3.1.2.6 Bus Interface

The system controller shall be the point of interface to the bus electrical platform and the preferred communication path shall be serial data (J-1939, CAN). The system controller shall also be capable of communication to a variety of on-board device controllers, including but not limited to, ABS, engine, and other.

3.1.2.7 Reserved

3.1.2.8 Reserved

3.1.2.9 Operator Interface

Operator interface shall include selection of desired direction, throttle, and visual indicators of faults. Alarms requiring immediate or indicating planned automatic shutdown shall also be audible. All interfaces shall be ergonomically designed.

3.1.2.10 Energy Storage

The traction energy storage system shall be composed of nickel-metal-hydride or lithium ion batteries along with associated power electronics interface and controls, diagnostic systems, and environment controls of a commercial design for transit duty. This system shall be designed to provide a load-leveling function in the hybrid propulsion system. The energy storage devices used, and their arrangement shall be selected and sized to meet bus performance specifications and design goals, including: reduced vehicle exhaust emissions, improved vehicle fuel economy, long cycle life, low life-cycle cost, safety, maintainability, durability, and simple, robust diagnostics.

The energy storage system shall be roof mounted.

The storage system shall include a management system to monitor and control the operating conditions within each energy storage system module, including voltage, current, and temperature. This system shall include over-current and over-temperature protection features that disconnect current to and from the energy storage modules in the event of an over-temperature or over-current condition and also derates performance (limits current) in the case of an over temperature condition.

3.1.2.11 Thermal Management

If necessary to maintain thermal management of the energy storage system under the ambient operating conditions required herein, the energy storage system shall be furnished with a self-contained, hybrid drive manufacturer approved, refrigerated cooling system dedicated only to the thermal management of the energy storage system. A warning light shall be supplied on the driver's panel indicating abnormal operational conditions (i.e.: excessive battery compartment temperature, loss of airflow, loss of power) within the energy storage device cooling system. This cooling system shall be rated to provide the required thermal management under the ambient operating conditions required herein and shall be manufactured and installed in compliance with the specifications and requirements of the hybrid drive manufacturer.

3.1.2.12 Electrical Safety System

A high voltage interlock monitoring and activation system shall be used to determine intentional or unintentional opening of high voltage circuits or enclosures and automatically neutralize all

devices capable of or producing high voltage power. This system shall be fully compliant with SAE J1673, and other applicable standards, and shall be active when the master run switch is in the OFF/CL/ID position and shall monitor all components and associated cabling within the hybrid system. When the master run switch is in the RUN/NITE position, the monitoring system shall provide an audible and visual warning on the operator's panel and provide for automatic shutdown of the high voltage system if a shutdown event occurs. The high voltage system includes all bus platforms operating above 50 VDC or 25 VAC and any variety of modulated currents to/from power electronic converters.
In addition, an electrical isolation fault detecting system shall monitor the level of isolation between the high voltage system (positive and negative) and the bus chassis and be fully compliant with SAE J1766, and all other applicable standards. Further, the vehicle's designed safety characteristics shall be supplemented by the use of properly labeled and located safety tags and plates in accordance to SAE J1654 and J1673.

3.1.2.13 Energy Storage System Battery Charging

Charging shall come from on-board or regenerative charging only. The propulsion system shall be designed so that external charging is not necessary. The energy storage unit shall be isolated from the passenger compartment so as to prevent any intrusion of liquids or gasses relating to the energy storage unit into the passenger compartment.

3.1.2.14 Diesel Engine

Each bus vehicle shall be powered by an in-line 6 cylinder diesel engine with a minimum rating of 280 BHP and meeting the requirements below.

The engine shall be in compliance with US EPA requirements for transit buses (see section 3.54 Warranty Requirements).

The engine shall be furnished with electronic controls and diagnostic system to protect the engine from serious damage due to high temperatures, or loss of lubrication or coolant.

- Piping or hoses containing fuel, oil, and other flammable liquids shall not be routed through wheel-housings or bundled with electric wires;

- Provisions for monitoring engine oil pressure and coolant temperature shall be provided in the engine compartment for ease of maintenance. All oil pressure and coolant temperature sensors must not have Teflon tape or any other liquid or paste sealer on the threads which may cause electrical resistance;

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The engine control system shall protect the engine against progressive damage. The system shall monitor conditions critical for safe operation and automatically derate power and/or speed and initiate engine shutdown as needed. An on-board diagnostic system shall trigger a visual and audible alarm to the operator whenever the engine control system detects a malfunction and the engine protection system is activated. Automatic shutdown shall occur when, at a minimum, parameters established for the following functions are exceeded:

Low coolant level
High coolant temperature
Low oil pressure

3.1.2.15 Engine Compartment

The engine compartment shall be furnished with the following:

- All fluid locations accessible with standard funnels, pour spouts, and automatic dispensing equipment.
- Engine compartment lights, controlled by a manual switch located at the rear of the compartment and accessible from the ground.
- Ready access to engine compartment for servicing and routine maintenance of engine and engine components.
- Hinged engine compartment cover(s) or door(s) with a positive hold open device for servicing. Taillight, stoplight, etc. wires running to hinged doors from the body shall have a flex loop, secured at both ends, to allow for normal opening of cover without damaging wires. A positive ground wire shall be used, not the hinges, for grounding components.
- The engine dipstick shall be routed to and easily accessible from the rear of the bus.
- The engine compartment, including the transmission housing area, shall be furnished with thermal and sound insulation applied to the top and sides of the engine/transmission compartment to minimize sound and heat transfer to the passenger side of the compartment.

3.1.2.16 Engine High-Idle Speed Control

An engine OEM high-idle speed control shall be furnished and meet the following requirements:

- Operate the engine at 950-1100±100 RPM RPM to sustain the bus's total continuous electrical load and maximum heating/air conditioning output.
- Operate only when engaged by a driver operated two-way switch mounted on the dash or side console and activated only when switched to the "ON" position,

the parking brakes are applied, the transmission is in “NEUTRAL” or “PARK”. The high idle speed shall be attained automatically by the engine ECM high-idle control with no manual intervention.

- The high-idle shall disengage when the transmission is placed in gear. The transmission shall not be allowed to engage in gear until the engine has returned to its low-idle speed setting.

3.1.2.17 Alternator

If low voltage 24/28 volt DC is furnished by the hybrid drive system in lieu of a separate, mechanically driven, alternator, the following alternator is not required.

If low voltage 24/28 volt DC is not furnished by the hybrid drive system, the alternator shall be a 24 Volt Leese-Neville, Niehoff, or Delco...
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3.1.2.18 Engine Cooling System

As a result of potential high relative altitude, temperature/seasonal extremes, and operating reliability demands, ordinary water-based cooling systems, which offer marginal performance in lower altitude, warmer climates are not acceptable. The contractor shall provide verifiable documentation from the proposed engine manufacture that the proposed cooling system has been designed and installed according to the engine manufacturer's recommendations demonstrating that it will meet performance requirements as defined in this specification.

As evidenced by the engine manufacturer’s application engineering acceptance tests, the cooling system shall maintain the engine coolant at a temperature not exceeding the engine OEM requirements when loaded to GVW and operated at an altitude of 7,000 feet above sea level in an ambient air temperature of not less than 109 degrees F.

Cooling system coolant shall be a 50/50 low silicate, ethylene glycol coolant mixture. The Government requires de-ionized water or soft water versus regular (hard) tap water to be mixed with the antifreeze. A coolant recovery tank shall also be provided which can be visually verified to determine the level of the fluid in the tank. In addition, the proposed radiator air inlet area shall be sufficiently large and unrestricted. If utilized, a proposed radiator door shall not have bumper guards, louvers, or thick screen(s) that restricts air from entering the radiator.

Engine manufacturer supplied hoses are acceptable. Any hose that is near (within 12 inches) of the radiant heat of the turbo charger and exhaust manifold will require a shield to prevent the hose from heat deterioration. A shield may be a split rubber hose covering the coolant hose and held in place with metal clamps. Split plastic loom is not acceptable.

The contractor shall provide a high capacity, high efficiency, thermostatically controlled coolant fan. The engine charge air cooling system shall provide maximum air intake temperature reduction with minimal pressure loss. The charge air radiator shall be sized and positioned to meet engine manufacturer's requirements. The charge air radiator shall not be stacked ahead or behind the engine radiator. Air ducting and fittings shall be protected against heat sources, and shall be configured to minimize restrictions and maintain sealing integrity.

All low points in the cooling system shall be equipped with drain cocks. All high points in the cooling system shall be vented from the surge tank.

Coolant and heater hoses shall be chemically resistant silicon or EPDM hoses utilizing constant tension type hose clamps.

All hoses shall be as short as possible but shall not be stressed.
All fluid lines shall be rigidly supported with rubber covered, corrosion resistant, metal hanger clamps to prevent chafing damage, fatigue failures, and tension strain a maximum of every 18".
All fluid lines shall be rigidly supported with rubber covered, corrosion resistant, metal hanger clamps to prevent chafing damage, fatigue failures, and tension strain a maximum of every 35".

3.1.2.19 Engine Coolant Heater

The engine shall be furnished with one 750 watt block coolant heater, with electrical cord(s) enclosed in a split nylon loom. The heater(s) shall be connected to weather proof plug(s). To help prevent corrosion on the plug prongs, a suitable anti-corrosion spray shall be applied to the plug connection (tangs). The block heater cavity shall be filled with appropriate dielectric grease before screwing the electrical cord onto the block heater. If a block heater is mounted in a coolant pipe/hose, the block heater shall have a separate ground wire to the engine block.

3.1.2.20 Oil Filter

The engine shall be furnished with an engine OEM approved oil filter.

3.1.2.21 Power Steering

A power steering system shall be provided. The steering gear shall be either axle-mounted or frame-mounted. The steering column shall have telescoping and tilt column adjustments. The steering columns shall be an integral type with the number and length of flexible lines minimized.

Fatigue life of all mechanical steering components shall exceed the service life of the bus. No element of the mechanical steering system shall fail before suspension system components when one of the tires strikes a severe road hazard. The mechanical steering system shall be considered as part of the basic body structure.

Turning effort:
Steering torque applied by the operator shall not exceed 10 foot-pounds with the front wheels straight ahead to turned 10 degrees. Steering torque may increase to 70 foot-pounds when the wheels are approaching the steering stops. Steering effort shall be measured with the bus at GVWR, stopped with the brakes released, and the engine at normal idling speed on clean, dry, level, commercial asphalt pavement and tires inflated to recommended pressure. Power steering failure shall not result in loss of steering control. With the bus in operation, the steering effort shall not exceed 55 pounds at the steering wheel rim, and perceived free play in the steering system shall not materially increase as a result of power assist failure. Gearing shall require no more than 7 turns of the steering wheel lock-to-lock.

Caster angle shall be set to provide a tendency for the return of the front wheels to the straight position with minimal assistance from the operator.

Turning Radius:
The outside body corner turning radius shall not exceed 44±1.0 feet with the bus at seated load weight (SLW).

3.1.2.22 Air Compressor

An engine driven air compressor and air system shall be furnished that meets the following requirements:

- Minimum 18.0 CFM
- Engine flange mounted and gear driven

See section 3.7 for air system requirements

Regardless of the system's air pressure, idle up to the rated engine speed shall be available to the operator with the transmission in neutral and the parking brake applied

3.1.2.23 Air Filter

The air filter shall be dry type, single or two-stage, engine OEM approved air cleaner with service indicator mounted on the dashboard of the bus or in the engine compartment in the engine air intake piping. If mounted in the engine compartment, the service indicator shall be easily observable by maintenance personnel standing on the ground.

3.1.3 Reserved

3.1.4 Fuel Tank(s)

The fuel tank(s) shall be equipped with an external, square or hex head, stainless steel drain plug. It shall be at least a 3/8-inch size and shall be located at the lowest point of the tank(s). The tank(s) shall be baffled internally to prevent fuel-sloshing noise regardless of fill level. The baffles or fuel pickup location shall assure continuous full power operation on a 6 percent upgrade for 15 minutes starting with no more than 25 gallons of fuel over the unusable amount in the tank(s). The bus shall operate at idle on a 6 percent downgrade for 30 minutes starting with no more than 10 gallons of fuel over the unusable amount in the tank(s).

The fuel tank(s) shall be securely mounted to the bus to prevent movement during bus maneuvers.

The capacity, date of manufacture, manufacturer name, location of manufacture, and certification of compliance to Federal Motor Carrier Safety Regulation shall be permanently marked on the fuel tank(s). The markings shall be readily visible and shall not be covered with an undercoating material.

3.1.4.1 Fuel Capacity

The fuel tank(s) shall have a total minimum working capacity of 95 gallons.

3.2 Fire Detection/Suppression System

The contractor shall furnish a fully automatic and integrated Amerex Vehicle Safety Net fire detection and dry chemical (ABC rated) fire suppression system consisting of the following:

- Amerex V25 fire detection and suppression system with a minimum of three (3) detectors and four (4) discharge nozzles. A manual actuator shall be furnished at the driver’s station

The system, integration, and installation shall be First Article Approved by Amerex Corporation.

3.3 Fluid Lines, Fittings and Clamps, and Charge Air Pipework

All fluid lines and air pipework shall be rigidly supported to prevent chafing damage, fatigue failures, and tension strain. All support clamps shall be rubber covered to prevent chafing and cutting.

Cooling piping shall be stainless steel or brass tubing.

Fuel, oil, and hydraulic lines shall be compatible with the fluid they carry. The lines shall also be compatible with potentially damaging elements of the surrounding environment including heat. Lines shall be capable of withstanding maximum system pressures.

Flexible fuel and oil lines shall be kept at a minimum and shall be as short as practicable. Flexible lines shall be routed or shielded so that failure of a line shall not allow fuel or oil to spray or drain onto any component operable above the auto-ignition temperature of the fluid.

Flexible hoses and fluid lines shall be individually supported and shall not touch one another or any part of the bus.

Fuel lines shall be rated and sized to prevent freezing and plugging due to condensation and/or fuel gelling in extreme winter. Flexible hoses shall be in conformance with SAE Standard J100R5 in SAE J517.

Charge air pipework and fittings shall be designed to minimize air restrictions and leaks. Pipework shall be as short as possible and the number of bends minimized. Bend elbows shall have smooth contours and shall be maximized to meet the pressure drop and temperature rise requirements of the engine manufacturers. The cross section of all charge air pipework shall not be less than the cross section of the intake manifold inlet. Any changes in pipework diameter shall be gradual to ensure a smooth passage of air and to minimize restrictions. Pipework shall be routed away from exhaust manifolds and other heat sources, and shall be insulated and shielded from these sources to protect inlet air from heat radiation.
Charge air pipework shall be seamless, constructed of stainless steel, aluminum, or aluminized steel. Connections between all charge air pipework sections shall be sealed with a short Section of reinforced hose and stainless steel, constant tension clamps that provide a complete 360 degree seal.

3.4 Suspension

3.4.1 General

All axle suspensions shall be of the pneumatic type and shall have a load rating equal to or exceeding that of the axles. The basic elements of the suspension system shall last the life of the bus without major overhaul or replacement. Suspension beams, weldments, and structural members shall be considered as parts of the basic body structure. Items such as bushings and air springs shall be easily and quickly replaceable by a 3M mechanic in 30 minutes or less. Suspension pivots shall be replaceable. Bushings shall be permanently lubricated or require no lubrication. Adjustment points shall be minimized and shall not be subject to a loss of adjustment in service.
Front and rear axle sway bars may be utilized if necessary.

3.4.2 Damping

Vertical damping of the suspension system shall be accomplished by hydraulic shock absorbers mounted to the suspension arms or axles and attached to an appropriate location on the chassis. Damping shall be sufficient to control coach motion to 3 cycles or less after hitting road perturbations. Shock absorbers shall maintain their effectiveness for at least 50,000 miles of the service life of the bus. Each unit shall be replaceable by a 2M mechanic in less than 1 hour.

3.4.3 Lubrication

All elements of steering, suspension, and drive systems requiring scheduled lubrication shall be provided with grease fittings conforming to SAE Standard J534. These fittings shall be located for ease of inspection, and shall be accessible with a standard grease gun without flexible hose end from a pit or with a bus on a hoist. Each element requiring lubrication shall have its own grease fitting with relief path. Lubricant specified shall be standard for all elements on the bus serviced by standard fittings.

3.4.4 Travel

The suspension system shall permit a minimum wheel travel of 3 inches jounce-upward travel of a wheel when the bus hits a bump (higher than street surface), and 3 inches rebound-downward travel when the bus comes off a bump and the wheels fall relative to the body. Elastomeric bumpers shall be provided at the limit of jounce travel. Rebound travel may be limited by elastomeric bumpers or hydraulically within the shock absorbers. Suspensions shall incorporate appropriate devices for automatic height control so that regardless of load the bus height relative to the centerline of the wheels does not change more than ± ½ inch at any point.

3.4.5 Kneeling

A full front kneeling system shall lower the entrance(s) of the bus a minimum of 2.5 inches during loading or unloading operations regardless of load up to GVWR, measured at the longitudinal centerline of the entrance door(s), by the driver using a three position, spring loaded to center switch. Downward direction will lower the bus. If the rear door platform height, in an unkneeled position, is no greater than 14±1.0 inches from the ground, there is no requirement for rear kneeling. Otherwise, a right rear kneeling system shall be furnished. Release of switch at anytime will completely stop lowering motion and hold height of the bus at that position. Upward direction of the switch will allow the system to go to floor height without the driver having to hold the switch up.

Brake and throttle interlocks shall prevent movement when the bus is kneeled. The kneeling control shall be disabled when the bus is in motion. The bus shall kneel at a maximum rate of 2 inches per second at essentially a constant rate. After kneeling, the bus shall rise within 4seconds to a height permitting the bus to resume service and shall rise to the correct operating height within 7 seconds regardless of load up to GVWR.

An indicator visible to the driver shall be illuminated until the bus is raised to a height adequate for safe street travel. An audible warning alarm will sound simultaneously with the operation of the kneeler to alert passengers and bystanders. A warning light mounted near the curbside of the front door, minimum 2.5 inches diameter, amber lens shall be provided that will blink when the kneel feature is activated. Kneeling shall not be operational while the wheelchair ramp is deployed or in operation.

3.5 Axles

Front and rear axles shall have a minimum load rating that exceeds the actual load on the axles when the bus is loaded to the GVWR, and shall operate for 300,000 miles without repairs. The front and rear axle ratings shall be the offeror's highest available rating to maximize the GVWR and the number of seated and standee passengers based on the gross load definition in Section 4.2-Definitions. The axle gearing shall be easily accessible for lubrication.

3.6 Brakes

The brakes shall be of the full-air internal expanding S-Cam type or disc type and shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.52 and FMVSS 571.121. Brake linings shall have extra service transit linings. Brake linings/pads shall be fabricated of non-asbestos material.

Brakes shall be self-adjusting. Any metallic air brake tubing shall comply with SAE J1149.

3.6.1 Reserved

3.6.2 Reserved

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3.6.3 Reserved

3.6.4 Parking Brake

The parking brake shall be a spring-operated system, actuated by a valve that exhausts compressed air to apply the brakes. The parking brake may be manually enabled when the air pressure is at the operating level per FMVSS 121. An emergency brake release shall be provided to release the brakes in the event of automatic emergency brake application. The parking brake valve button shall pop out when air pressure drops below requirements of FMVSS 121. The driver shall be able to manually depress and hold down the emergency brake release valve to release the brakes and maneuver the bus to safety. Once the operator releases the emergency brake release valve, the brakes shall engage to hold the bus in place.

3.7 Air System

3.7.1 General

The bus air system shall operate the air-powered accessories and the braking system with reserve capacity. A new bus shall not leak down more than 5 psi as indicted on the instrument panel mounted air gauges, within 15 minutes from the point of governor cut-off.

Provision shall be made to apply shop air to the bus air systems using a standard tire inflation type valve. A quick disconnect fitting shall be easily accessible and located in the engine compartment and near the front bumper area for towing. Retained caps, where approved for use by the OEM, shall be installed to protect fittings against dirt and moisture when not in use. Air for the compressor shall be filtered through the main engine air cleaner system. The air system shall be protected by a pressure relief valve and shall be equipped with check valve and pressure protection valves to assure partial operation in case of line failures.

3.7.2 Air Lines and Fittings

Air lines subjected to temperatures over 200 Deg F shall be SAE100R14A PTFE, stainless steel braid covered hose, and all other air lines shall be nylon tubing manufactured in accordance with SAE J844 with performance meeting the requirements of SAE J1131. Nylon tubing shall be installed in accordance with the following color-coding standards:

- Green: Primary brakes and supply;
- Red: Secondary brakes;
- Brown: Parking brake;
- Yellow: Compressor governingSuspension;
- Black: Accessories.

Nylon lines may be grouped and shall be continuously supported and prevented from any movement, flexing, tension strain, and vibration. They shall not touch one another or any part of the bus except for the supporting grommets. Flexible lines shall be supported at 30” intervals or less. Grommets of bulkhead fittings shall protect the air lines at all points where they pass through understructure components.

3.7.3 Air Reservoirs

All air reservoirs shall meet the requirements of FMVSS Standard 121 and SAE Standard J10. The wet or “ping” tank shall be furnished with automatic, heated, drain valve. All other air reservoirs shall be equipped with manual drain valves, unless mounting locations allow draining via the wet tank. The manual drain valves shall be located such that they may be drained from the ground at the side of the bus. The system shall be furnished with a pressure relief apparatus equipped with check valve, Major structural members shall be provided to protect these valves from road hazards. Reservoirs shall be sloped toward the drain valve to assure that condensate is eliminated by the valve. All air reservoirs and tanks shall have a non-corrosive treatment applied to them prior to installation on the bus vehicle (subject to Government review and approval).

3.7.4 Air Dryer

An air dryer with replaceable desiccant cartridge shall be furnished on all air systems and installed per compressor and air dryer manufacturer’s recommendations.

3.8 Hydraulic Systems

Any accessory may be driven hydraulically. The hydraulic system shall demonstrate a mean time between repairs in excess of 50,000 miles. Hydraulic system service tasks shall be minimized and scheduled no more frequently than those of other major bus systems. All elements of the hydraulic system shall be easily accessible for service or unit replacement.

3.8.1 Hydraulic systems shall meet the following requirements:

Hydraulic systems shall meet the following requirements:

- Hydraulic system hoses, fittings, pressures, and flow rates.

  Hydraulic hoses shall be rubber covered double wire braid reinforced and comply with SAE 100R2, type A or AT, or 100R9, type A or AT, or single wire braid reinforced hose complying with SAE 100R5 of SAE J517. The working pressure of the hose shall exceed the pressure setting of the relief valve. Hoses shall be sized such that the maximum velocity of hydraulic fluid in the hose does not exceed the following:
  A. Fluid velocity in suction lines shall not exceed 4 ft./sec.
  B. Fluid velocity in discharge lines shall not exceed 25 ft./sec

- All hoses shall be installed in accordance with the requirements and recommendations of SAE J1273
- System working pressure shall not exceed 3500 PSI.
- High pressure hydraulic hose fittings shall comply with the requirements of SAE J516 for permanently attached (crimped) fittings with JIC 37° flare. Field replaceable type fittings are acceptable on low pressure return and suction hoses. Forged steel hydraulic adapters shall be used. Cast steel fittings are acceptable only in low pressure circuits, such as suction lines.

  Any other hydraulic system shall be designed with adequate cooling (reservoir dwell time or auxiliary cooler) such that the maximum oil temperature does not exceed 200 Deg. F.

3.9 Odometer, Hub Mounted

Each bus shall be supplied with a hubodometer.

3.10 Tires and Wheels

3.10.1 Tires

Tires shall be suitable for the conditions of transit service and sustained operation at the maximum speed capability of the bus. Load on any tire at GVWR shall not exceed the tire supplier's rating.
The Contractor shall provide "plain" valve system caps with each mounted tire. No valve stem tool will be permitted on the valve stem cap.

3.10.2 Wheels

Wheels shall be drop forged aluminum, hub piloted, and designed to meet all operational parameters and resist rim flange wear. The wheels shall be polished on both sides. All wheels shall be interchangeable and shall be removable without a puller. Wheels shall be of the ventilated type with 10 studs, radial approved, reverse mount, and made for tubeless tires. Wheels shall be compatible with tires in size and load-carrying capacity. Front wheels and tires shall be balanced as an assembly per SAE J1986.

Front wheels and tires shall be spin balanced as an assembly using appropriate weights. Not more than sixteen (16) ounces of weight per wheel may be used to balance the tire.
The contractor shall furnish one fully balanced spare wheel/tire assembly, shipped loose.

3.11 Exhaust

The exhaust system shall be constructed of stainless steel tubing and stainless steel bellows connectors. The exhaust gases and waste heat shall be discharged from the roadside rear corner of the bus vehicle roof. The exhaust pipe shall be of sufficient height to prevent exhaust gases and waste heat from discoloring or causing heat deformation to the bus roof. The entire exhaust system shall be adequately shielded to prevent heat damage to any bus component. The exhaust outlet shall be of curve designed to prevent rain, snow, or water generated from high-pressure washing systems from entering into the exhaust pipe and causing damage to the catalyst.

3.12 Service Requirements

3.12.1 General

The engine shall be arranged so that accessibility for all routine maintenance is assured. Two 3M mechanics shall be able to remove and replace the engine and transmission assembly in less than twelve (12) total combined man-hours.

The muffler, exhaust system, air cleaner, air compressor, starter, alternator, radiator, cooling system surge tank, all accessories, and/or any other component requiring service or replacement shall be mounted in an accessible location. Each bus shall be designed to facilitate the disassembly, re-assembly, servicing or maintenance by use of tools and items that are normally available as commercial standard items. Any special tools required must obtain the approval of the Government (see Section 3.56).

3.12.2 Fillers

Engine oil and coolant filler caps shall be properly labeled. All fillers shall be accessible with standard funnels, pour spouts, and automatic dispensing equipment.

3.12.3 Accessories

Wherever possible, all engine driven accessories, such as alternator, power steering pump, and air compressor, shall be modular and unit-mounted for quick removal and repair. Accessory drive systems shall operate without failure or unscheduled adjustment for 50,000 miles on the design operating profile. These accessories shall be driven at speeds sufficient to assure adequate system performance during extended periods of idle and low route speeds typical of transit operation. All engine mounted accessories shall be driven by poly-V type belts, and non-engine mounted, engine driven accessories, such as the air conditioner compressor, shall be driven by Powerband type V-belts. Belt guards shall be provided for all belts.

3.13 Body

3.13.1 General

3.13.2 Design

The bus shall be of integral body and understructure construction and shall have a clean, smooth, simple design and uniform appearance. The exterior and body features, including grilles and louvers, shall ideally be shaped to facilitate cleaning by automatic bus washers without causing damage to either the bus vehicle body or the washer brushes. Water and dirt shall not be retained in or on any body feature to freeze or bleed out onto the bus after leaving the washer. The body and windows shall be sealed to prevent leaking of air, dust, or water under normal operating conditions and during cleaning in automatic bus washers for the service life of the bus. In addition, as part of Contractor proposals, an exterior vehicle rendering of the proposed vehicle shall be included.

Accumulation on any windows of the bus of spray and splash generated by the bus wheels on a wet road shall be minimized. Heated exterior rear view mirrors shall be furnished.

3.13.3 Materials and Construction

All body materials shall be selected and the body fabricated for simplified replacement and repair, to reduce maintenance, extend durability, and provide consistency of appearance throughout the service life of the bus. Add-on devices and trim, where necessary, shall be minimized and integrated into the basic design. All exterior panels shall be attached to the vertical wall studs with either industrial adhesive or buck rivets. Sheet metal screws in any location are not acceptable.

3.13.4 Crashworthiness

The bus vehicle body and roof structure shall withstand a static load equal to 150 percent of the curb weight evenly distributed on the roof with no greater than a 6-inch reduction in any interior dimension. Under such a load, windows shall not open and will remain in place.

At any point (excluding doorways), the bus vehicle shall withstand a 25-mph impact by a 4,000-pound automobile along either side of the bus with no more than 3 inches of permanent structural damage at seated passenger hip height. Sharp edges or protrusions in the bus vehicle interior shall not result from this impact.

Exterior panels located below 35 inches from ground level shall withstand a static load of 2,000 pounds applied perpendicular to the bus by a pad no larger than 5 inches square. Deformation that prevents installation of new exterior panels to restore the original appearance of the bus vehicle shall not result from the application of this load.

The contractor shall furnish to the Government evidence of the crashworthiness capabilities of the proposed bus (see Section 3.56).

3.13.5 Fire Protection

The passenger and engine compartment shall be separated by fire-resistant bulkheads. The engine compartment shall include areas where the engine and exhaust system are housed. This bulkhead shall preclude or retard propagation of an engine compartment fire into the passenger compartment and shall be in accordance with the Recommended Fire Safety Practices defined in FMVSS 302. Only necessary openings shall be allowed in the bulkhead, and these shall be fire-resistant. Any passageways for the climate control system air shall be separated from the engine compartment by fire-resistant material. Piping through the bulkhead shall have fire-resistant fittings sealed at the bulkhead. Wiring may pass through the bulkhead only if connectors or other means are provided to prevent or retard fire propagation through the bulkhead. Engine access panels in the bulkhead shall be fabricated of fire-resistant material and secured with fire-resistant fasteners. These panels, their fasteners and the bulkhead shall be constructed and reinforced to minimize warping of the panels during a fire that will compromise the integrity of the bulkhead.

A certified, standard size ABC rated dry chemical, carbon dioxide, minimum 2.3 kg (5 lb.) unit, with dial type indicator, in quick-release brackets, shall be installed on each bus vehicle by the contractor. Proposed placement shall permit effective access by the bus vehicle operator and passengers in the event of fire while minimizing potential for vandalism or inadvertent tampering. Proposed placement within the bus vehicle shall be defined by the contractor and will be approved by the Government. In addition, a fully automatic fire detection/suppression system shall be provided for the engine compartment (see Section 3.2).

3.13.6 Resonance and Vibration

To minimize audible, visible, or sensible resonant vibrations during normal service, all structure, body, and panel-bending mode frequencies, including vertical, lateral, and torsional modes, shall be sufficiently removed from all primary excitation frequencies.

3.13.7 Corrosion Protection

The bus chassis structure shall resist corrosion or deterioration from atmospheric conditions and road salts for a period of 12 years or 500,000 miles which ever comes first. It shall maintain structural integrity and maintain nearly original appearance throughout its service life, provided that it is maintained by the Government in accordance with the procedures specified in the contractor's service manual. With the exception of periodically inspecting the visible coatings applied to prevent corrosion and reapplying these coatings in limited spots, the contractor shall not require the complete reapplication of corrosion compounds over the life of the bus.

All exposed surfaces and the interior surfaces of tubing and other enclosed members shall be corrosion resistant. All sub-assembly components requiring painting shall be primed and painted before installing into their next assembly.

All materials that are not inherently corrosion resistant shall be protected with corrosion-resistant coatings. All joints and connections of dissimilar metals shall be corrosion-resistant and shall be protected from galvanic corrosion. Representative samples of all materials and connections shall withstand a 2-week (336-hour) salt spray test in accordance with ASTM Procedure B-117 with no structural detrimental effects to normally visible surfaces, and no weight loss of over 1 percent.

3.13.8 Insulation

Polystyrene sheet or foam insulation shall be installed to fill cavities in the inner and outer walls and roof. Walls and ceiling shall have an "R" value of not less than R-5.

The vehicle body shall be thoroughly sealed so that, during normal operations, the operator or passengers cannot feel drafts with the passenger doors closed.

3.14 Structure

3.14.1 Strength and Fatigue Life

The basic structure shall be designed so that fatigue damage will not occur during the service life of the bus. The vehicle structure shall withstand impact and inertial loads due to street travel as representative in transit service throughout its service life, without permanent deformation or damage.

3.14.2 Hoisting

The bus axles and/or jacking plates shall accommodate the lifting pads of a 2-post hoist system. Jacking plates, if used as hoisting pads, shall be designed to prevent the bus from falling off the hoist. Other pads or the bus structure shall support the bus on jack stands independent of the hoist. The hoisting and jacking of a bus vehicle shall not create an over-capacity condition to any of the axles or air bags that may cause damage.

3.14.3 Jacking

It shall be possible to safely jack up the bus, at curb weight, with a common 10-ton floor jack with or without special adapter, when a tire or dual wheel set is completely flat and the bus is on a level, hard surface (without crawling under any portion of the bus). Jacking from a single point shall permit raising the bus sufficiently high to remove and reinstall a wheel and tire assembly. Jacking pads located on the axle or suspension near the wheels shall permit easy and safe jacking with the flat tire or dual wheel set on a run-up block not wider than a single tire. Jacking and changing any one tire shall be completed by a serviceman in less than 30 minutes from the time a bus is approached. The bus shall withstand such jacking at any one or any combination of wheel locations without permanent deformation or damage.

Jacking pads shall be painted safety yellow or orange for easy identification.

3.14.4 Towing

Towing devices shall be provided on the front end only of the bus. Each towing device shall withstand, without deformation, tension loads up to 1.2 times the curb weight of the bus within 20 degrees of the longitudinal axis of the bus.

Front towing devices shall allow attachment of adapters for a rigid tow bar and shall permit lifting and towing of the bus, at curb weight, until the front wheels are off the ground.

3.14.5 Flooring

3.14.5.1 Design

The floor shall be a continuous flat plane, unless interior steps are provided. Where the floor meets the walls of the bus, as well as other vertical surfaces, such as platform risers and wheel wells, the surface edges shall completely sealed with an appropriate sealant. The floor shall be sloped within ADA limitations in door entry thresholds to facilitate adequate drainage.

3.14.5.2 Construction

The floor shall consist of the subfloor and the floor covering. The floor, as assembled, including the sealer, attachments and covering shall be waterproof, non-hygroscopic, and resistant to mold growth. The subfloor shall be resistant to the effects of moisture, including decay (dry rot) and impervious to wood destroying insects such as termites. Plywood, if used, shall be of a thickness adequate to support the design loads, manufactured with exterior glue of Group I Western panels as defined in PS 1-95 (APA designation) and of a grade that is manufactured with a solid face and back. The edges of the floor shall not be exposed to tire water spray. The rear step/transition area between the upper and lower decks shall be composite flooring.

3.14.5.3 Strength

The floor deck may be integral with the basic structure or mounted on the structure securely to prevent chafing or horizontal movement and designed to last the life of the bus. Sheet metal screws shall not be used to retain the floor. The use of adhesives to secure the floor to the structure shall be allowed only in combination with the use of bolt or screw fasteners and its effectiveness shall last throughout life of the coach. All floor fasteners shall be secured and protected from corrosion for the service life of the bus. The floor deck shall be reinforced as needed to support passenger loads. At GVWR, the floor shall have an elastic deflection of no more than 0.60 inches from the normal plane. The floor shall withstand the application of 2.5 times gross load weight without permanent detrimental deformation. Floor, with coverings applied, shall withstand a static load of at least 150 pounds applied through the flat end of a 2 inch-diameter rod, with 1/32-inch radius, without permanent visible deformation.

3.15 Wheel Housings

3.15.1 Design

Sufficient clearance and air circulation shall be provided around the tires, wheels, and brakes to preclude overheating when the bus is operating on the design operating profile.

Wheel housings shall be adequately reinforced where seat pedestals are installed to ensure satisfactory performance over the life of vehicle in accordance with Section 3.14.1.

3.15.2 Construction

Wheel housings shall be constructed of stainless steel or fiberglass. Wheel housings, as installed and trimmed, shall withstand impacts of a two inch steel ball with at least 200 lbs-ft. of energy. The housings shall have sufficient insulation to minimize tire and road noise.

Wheel housings shall be designed to maximize the interior aisle width and preclude water from being trapped between the flooring and the top of the wheel housing. Wheel housings shall be equipped with liners and/or sealed to prevent water and dirt from entering or collecting in the frame or structure, causing corrosion.

3.15.3 Clearance

Sufficient clearance and air circulation shall be provided around tires, wheels, and brakes to prevent overheating when the bus is operating. Wheels and tires shall be removable when a bus vehicle is jacked by the axle or suspension, even with the air bags depleted. Tire chain clearance shall be provided on all driven wheels in accordance with SAE J1232-1980 Class S.

3.16 Exterior

3.16.1 Strength and Installation

Exterior panels may be structural components. Exterior surface panels shall be installed or retained with minimal visible rivets or other acceptable attachment method(s). Any penetrations made into bus structural components shall be sealed to prevent water from entering the inside of the component.

3.16.2 Rain Gutters

Rain gutters shall be provided to prevent water flowing from the roof onto side windows, doors, and exterior mirrors. When the bus vehicle accelerates or decelerates, the gutters shall not drain onto the windshield, or operator's side window, or into the door boarding area. Cross sections of the gutters shall be adequate for proper operation and gutters shall extend a minimum of two inches beyond openings.

3.16.3 License Plates

Provisions shall be made to mount standard size U.S. license plates per SAE J686 on the front and rear of the bus. To the extent practicable, license plates shall not allow a toehold or handhold for unauthorized riders.

3.17 Finish and Color

Bus finish may be composite with gelcoat, or painted. Unless a multi-color option is selected in the Options List described in Section 4.5 below, the exterior shall be the OEM's standard white. If other optional colors are specified, the roof shall remain the OEM standard white.

All exterior surfaces shall be smooth and free of wrinkles and dents. Exterior surfaces to be painted shall be properly prepared as required by the paint system supplier, prior to application of paint to assure a proper bond between the basic surface and successive coats of original paint for the service life of the bus. Drilled holes and cutouts in exterior surfaces shall be made prior to cleaning, priming and painting to prevent corrosion. Any paint shall be applied prior to installation of exterior lights, windows, mirrors and other items that are applied to the exterior of the bus. Body filler materials may be used for surface dressing, but not for repair of damaged or improperly fitted panels.

Depending upon the primer/topcoat system selected, painted surfaces shall be cleaned with solvent, chemical, or mechanical methods meeting the paint manufacturer's recommendations.

Paint used shall be a rust inhibitive primerand polyurethane topcoat or equal.

Paint or gelcoat shall be applied smoothly and evenly with the finished surface free of dirt and the following other imperfections:

A. Blisters or bubbles appearing in the topcoat film.
B. Chips, scratches, or gouges of the surface finish.
C. Cracks in the film.
D. Craters where finishfailed to cover due to surface contamination.
E. Overspray.
F. Peeling
G. Runs or sags from excessive flow and failure to adhere uniformly to the surface.
H. Chemical stains and water spots.
I. Orange peel

All dents, roughness, or other surface imperfections shall be corrected prior to the application of the priming coat. Primer and finish paint shall be supplied as a system by a single manufacturer. Preparation for finish application shall follow the finish manufacturer's recommendations. As a minimum, prior to finish application, surfaces shall be cleaned to remove all traces of contamination.

Finish shall be applied evenly, and the finished surface shall be free of dirt, runs, "orange peel" or other imperfections. At least two coats of paint shall be applied, with appropriate surface preparation between coats. Touch-up finish shall be identical in all respects to the original finish.

Prior to assembly, all low-alloy steel areas shall be painted with one coat of an approved primer followed by one coat of an approved sealer to prevent rusting.

3.17.1 Underfloor

Exposed underfloor steel surfaces, after installation of welded-on bracketry, shall be undercoated or primed and undercoated.

3.17.2 Equipment Enclosure(s)

The exterior of all equipment enclosures shall be finished. The interior of all equipment enclosures shall be finished.

3.17.3 Vehicle Body Exterior

All exterior finished surfaces shall be impervious to commercial cleaning agents. Finished surfaces shall resist damage by controlled applications of commonly used graffiti-removing chemicals. Except for periodic cleaning, exterior surfaces of the bus shall be maintenance-free, permanently colored and not require refinish/repaint for the life of the vehicle.
Colors and paint schemes shall be provided by the Government.

3.17.4 Roof

The roof shall be painted or gelcoat. If painted shall receive a minimum of a total of 4 mils of primer and topcoat as specified herein.

3.17.5 Exposed Piping, Conduits and Wireways

The contractor shall ensure that wireways, conduits, and piping that can be corroded shall receive a minimum of two coats of primer and two coats of an approved paint. This priming and painting can be accomplished either before or after installation of the item on the bus vehicle body.

3.17.6 Undercoating

The OEM’s standard undercoating material shall be applied according to the manufacturer's instructions. Undercoating thickness shall conform to the manufacturer's recommended practices.
The composition of the materials selected for the protective treatment system shall be such that treated members may be readily cut or removed using conventional cutting methods, such as burning or sawing, without creating any fumes hazardous to employees.

All exposed conduit, piping, and bracketry shall be protected with undercoating.

3.18 Fender Skirts

Features to minimize water spray from the bus in wet conditions shall be included in wheel housing design. Any fender skirts shall be unbreakable and easily replaceable. They shall be flexible if they extend beyond the allowable body width. Wheels and tires shall be removable with the fender skirts in place.

3.19 Splash Aprons

Splash aprons, composed of composition or rubberized fabric, shall be installed with each wheel and shall extend downward. Apron widths shall be no less than tire widths and extend downward to within 6" of the road surface (static). Splash aprons shall be bolted to the bus understructure. Splash aprons and their attachments shall be inherently weaker than the structure to which they are attached. The flexible portions of the splash aprons shall not be included in the road clearance measurements. Other splash aprons shall be installed where necessary to protect bus equipment.

3.20 Windshield Wipers and Washers

The bus shall be equipped with variable speed windshield wipers complying with SAE J198. If powered by compressed air, exhaust from the wiper motors shall be muffled or piped under the floor of the bus. Wiper sweep shall allow unobstructed viewing of exterior rear mirrors. If two window panes are provided, both wipers shall park along the center divider of the windshield. If a single pane is provided, the wipers shall park along the lower edge of the windshield so as not to obstruct the vision of the operator. Windshield wiper motors and mechanisms shall be easily accessible for repairs or service, mounted on mechanical fasteners, and removable as individual units from the exterior of the bus.

The windshield washer system shall deposit washing fluid on the windshield and, when used with the wipers, shall evenly and completely wet the entire wiped area.

Reservoir pumps, lines, and fittings shall be corrosion resistant, and the reservoir itself shall be translucent for easy determination of fluid level.

3.21 Service Compartments and Access Doors

Service compartments and access doors shall have stainless steel hinges that shall be used for access to the engine compartment and all auxiliary equipment compartments. Access openings shall be sized for easy performance of tasks within the compartment, including tool-operating space. Handles shall be constructed of non-corroding material and shall be flush with, or recessed into, the body contour and shall be sized to provide an adequate grip for opening. Springs and hinges shall be stainless steel.

Access doors shall be of rugged construction and shall maintain mechanical integrity and function under normal operations throughout the service life of the bus. They shall close flush with the body surface and shall be prevented from coming loose or opening during transit service or in bus washing operations. Access doors, when opened, shall not restrict access for servicing other components or systems. The curbside and roadside engine compartment access doors shall be mounted to horizontal hinges so that they will fold up and out of the way; no vertical hinges shall be permitted.

Access doors shall be retained in the opened and closed positions with over-center gas-filled springs, without the use of latches except for electrical and radio compartments. Doors smaller than 36 square inches shall be retained in the open and closed positions by over-center springs, gas springs, or magnetic latches.

The battery compartment shall be of stainless steel or other non-corrosive material, vented and self-drained, and prevent accumulation of debris on top of the batteries. It shall be accessible only from the outside of the bus. All components within the battery compartment, and the compartment itself, shall be protected from damage or corrosion from the electrolyte. The inside surface of the battery compartment’s access door shall be electrically insulated, as required, to prevent the battery or plus terminals from short-circuiting on the door.
The batteries shall be securely mounted on a stainless steel or equivalent non-corrosive tray that can accommodate the size and weight of the batteries. The battery tray shall pull out easily and properly support the batteries while they are being serviced. The tray shall allow each battery cell to be easily serviced and filled. A locking device shall retain the battery tray in the stowed position.

All components within the battery compartment, and the compartment itself, shall be protected from damage or corrosion from the electrolyte and gases emitted by the battery, and from snow, slush, salt spray, mud, etc. generated from environmental conditions outside the vehicle.

3.22 Bumpers

Bumpers shall provide impact protection for the front and rear of the bus. The bumpers shall not exceed allowable bus width. Bumper height shall be such that when one bus is parked behind another, a portion of the bumper faces will contact each other. Bumper material shall be corrosion-resistant and withstand repeated impacts of the specified loads without sustaining damage. Visible surfaces shall be black or color -coordinated with the bus exterior. These qualities shall be sustained throughout the service life of each bus.

For front bumpers, no part of each bus, including the bumper, shall be damaged as a result of a 5-mph impact of the bus at curb weight with a fixed, flat barrier perpendicular to the longitudinal centerline of the bus. The bumper shall return to its pre-impact shape. The energy absorption system of the bumper shall be independent of every power system of the bus and shall not require service or maintenance in normal operation during the service life of the bus.

For rear bumpers, no part of each bus, including the bumper, shall be damaged as a result of a 2-mph impact with a fixed, flat barrier perpendicular to the longitudinal centerline of the bus. The bumper shall return to its pre-impact shape.

The rear bumper or bumper extensions shall be shaped to preclude unauthorized riders standing on the bumper. The bumper extensions shall not hinder service and shall be faired into the body of each bus with no protrusions or sharp edges. The rear bumper shall be independent of all power systems of each bus and shall not require service or maintenance in normal operation during the service life of the bus.

Any flexible portion of the bumpers shall minimally increase the overall length of the bus vehicle.

3.23 Exterior Lighting

All exterior lights shall be designed to prevent entry and accumulation of moisture or dust, and each lamp shall be replaceable by a 2M mechanic within 5 minutes. Commercially available LED (Light Emitting Diode)-type lamps must be used on all exterior lights except headlights, headlights may also be LED type. Lights mounted on the engine compartment doors shall be protected from the impact shock of door opening and closing. Lamps, lenses and fixtures shall be interchangeable to the extent practicable.

Hazard lamps at the rear of the bus shall be visible from behind when the engine service doors are opened. Light lenses shall be designed and located to prevent damage when running the vehicle through an automatic bus washer. Lights located on the roof and sides (directionals) of the bus shall have protective shields or be of the low profile type to protect the lens against minor impacts.

Turn-signal lights shall be provided on both sides of the bus. Visible and audible warning shall inform following vehicles or pedestrians of reverse operation. Visible reverse operation warning shall conform to SAE Standard J593. Audible reverse operation warning shall conform to SAE Recommended Practice J994 Type C or D.

Lamps at the front and rear passenger doorways shall comply with ADA requirements and shall activate only when the doors open and shall illuminate the street surface. The lights may be positioned above or below the lower daylight opening of the windows and shall be shielded to protect passengers' eyes from glare.

Rear cornering illumination shall be furnished on the passenger side.

3.24 Furnishings and Interior Arrangement

All furnishing design, arrangement, installation, and detail drawings shall be provided to the Government for acceptance (see Section 3.56).

3.24.1 Windshield

The windshield shall permit an operator's field of view as referenced in SAE Recommended Practice J1050. The vertically upward view shall be a minimum of 15 degrees, measured above the horizontal and excluding any shaded band. The vertically downward view shall permit detection of an object 3-1/2 feet high no more than 2 feet in front of the bus. The horizontal view shall be a minimum of 90 degrees above the line of sight. Any binocular obscuration due to a center divider may be ignored when determining the 90-degree requirement, provided that the divider does not exceed a 3-degree angle in the operator's field of view. Windshield pillars shall not exceed 10 degrees of binocular obscuration. The windshield shall be designed and installed to minimize external glare as well as reflections from inside the bus.

The windshield shall be easily replaceable from the bus exterior by removing zip-locks from the windshield retaining moldings. Bonded-in-place windshield shall not be used. The windshield glazing material shall have a 1/4-inch or 6-mm nominal thickness laminated safety glass conforming to the requirements of FMVSS 205 and the Recommended Practices defined in SAE J673. The glazing material shall have single density tint. The upper portion of the windshield above the operator's field of view shall have a dark, shaded band with a minimum luminous transmittance of 6 percent when tested in accordance to ASTM D-1003.

3.24.2 Operator's Side Window

The operator's side window shall open sufficiently to permit the seated operator to easily adjust the street side outside rearview mirror. The operator's window in the open position shall not block the operator's view of the exterior mirror. The movable window section shall slide rearward in tracks, channels, and seals designed in a two section sash and last the service life of the bus. The operator's side window shall not be bonded in place and shall be easily replaceable. The glazing material shall have a uniform, single density, neutral tint. Side window shall be non-whistling.

The operator’s side window glazing material shall be of ¼” laminated safety glass conforming to the requirements of the Recommended Practices defined in SAE J673 and shall meet all FMVSS requirements.

3.24.3 Passenger Windows

3.24.3.1 Materials

Side windows glazing material shall use safety single density tint glass.

Minimum glazing requirements shall be as follows:
- Meet all FVMSS requirements
- Nominal thickness=0.25 inches
- Maximum luminous transmittance=44% gray (ASTM D-1003)

3.24.3.2 Dimensions

As a minimum, 8,000 square inches (total) of side vehicle window area (per vehicle side) shall be provided.

All side passenger windows, except windows in passenger doors and those smaller than 500 square inches, shall have window panels that are openable by passengers. Openable window panels shall be equipped with latches that secure the window in the closed position. Each side window shall be a transom style window, with the top openable portion opening inward.

All side windows shall be easily replaceable without disturbing adjacent windows and shall be mounted so that flexing or vibration from engine operation or normal road excitation is not apparent. The windows shall be designed and constructed to enable a 2M mechanic to remove and replace two windows within thirty (30) minutes.

3.24.4 Emergency Exits

The bus shall be provided with adequate exits for quick passenger escape during emergency conditions. With the exception of the last three windows, two (2) windows on the street side and one (1) window on the curb side (accessible by passengers) of each bus vehicle are to be emergency egress windows and shall comply with applicable codes and requirements and the best industry practices.

Passenger window emergency exits shall open outward to provide an emergency exit. A simple red latch (or equal subject to Government review and approval) shall be provided on all emergency egress windows. This latch shall not pinch a person's fingers or hands when operating. Each emergency window location shall be labeled (preferably close to the latch).

3.24.5 Doors

3.24.5.1 General

Two doorways shall be provided on the curb-side of each bus for passenger ingress and egress. The front doorway shall be forward of the front wheels. The rear doorway centerline shall be rearward to the point midway between the front door centerline and the rearmost seat back. Passenger doors and doorways shall comply with the applicable ADA requirements.
With the doors in any position, the operator's visibility of any mirror on each bus shall not be reduced. Doors are to be constructed to minimize vibratory noise and enhance simplified maintainability.

The front door shall be a Vapor Bus International slide-glide door and the rear door shall be a Vapor Bus International plug or slide-glide style door, furnished with Vapor Bus International's "CLASS"- Contactless Acoustic Sensing System.

3.24.5.2 Dimensions

Front and rear door width and height dimensions shall be in accordance with the successful offeror's drawing approved by the Government.

3.24.5.3 Materials

Structure of the doors, their attachments, inside and outside trim panels, and any mechanism exposed to the elements shall be durable and corrosion-resistant. Door panel construction shall be of corrosion-resistant metal or reinforced non-metallic composite materials. The doors, when fully opened, shall provide a firm support and shall not be damaged if used as an assist by passengers. Sloped grab handles shall be provided on the inside of all door panels.

All reasonable means shall be taken to reduce the weight of the door. The exterior of door panels shall be made of a material proven on similar buses and approved by the Government.

Door panels, operators, and associated drive components shall have sufficient strength and rigidity to sustain, without permanent deformation, a force of 198 lbs. applied to an area of 1 square foot located 2 inches from the leading edge and centered within the door height.

3.24.5.4 Door Glazing

The upper sections of both front and rear doors shall be glazed for no less than 45 percent of the respective door opening area of each section. The front door panel glazing material shall have a nominal ¼ inch thick laminated safety glass conforming to the requirements of MVSS 205 and the Recommended Practices defined in SAE J673, and meet or exceed the requirements of FMVSS 217.

3.24.5.5 Door Height (above pavement)

It shall be possible to open and close either passenger door when a bus loaded to GVWR is not knelt and parked with the tires touching a typical curb on a street sloping toward the curb so that the street side wheels are higher than the curb side wheels.

3.24.5.6 Door Sealing

The leading edges of each door panel shall form an acoustic and weather-tight seal. The edge material shall have sufficient stiffness to activate the sensitive edge control (if sensitive edge is used as a means of obstacle detection) and prevent the doors from locking when an obstacle (as defined above) is inserted between the panels.

The seal shall be readily replaceable without requiring the door panel to be removed from the vehicle. Doors shall be weather-stripped to seal against weather, odors, and running noise. Seals shall not result in friction loading on the panels.

Doors shall be approximately flush with the bus vehicle body exterior when closed. Door panels shall be sealed against rain and bus vehicle washing. Adequate drainage from the door panels and thresholds shall be provided.

3.24.5.7 Actuators

Door actuators shall be independently adjustable, either electrically or pneumatically driven, and controlled electrically. Actuators and the door mechanism shall be concealed from passengers but shall be easily accessible for servicing.

3.24.5.8 Emergency Operations

In the event of an emergency, after actuating an unlocking device at each door, it shall be possible to open the doors manually from inside the bus using a force of no more than 25 pounds. An unlocking device shall require two distinct actions to actuate and shall be clearly marked as an emergency-only device. Precise instructions for the emergency exit shall be posted near the device, and the door emergency unlocking device shall be accessible from the doorway areas. The door interlock throttle system shall return the engine to idle and the door interlock brake system shall apply to stop the bus when this emergency device is actuated.

In order to open manually, locked doors shall require a force of more than 100 pounds. Damage shall be limited to the bending of minor door linkage without resulting damage to the doors, engines, and critical vehicle mechanisms when the locked doors are manually forced to open.

3.25 Reserved

3.26 Interior Surfaces and Finishes

3.26.1 General

Interior surfaces below the lower edge of the side windows or windshield shall be shaped so that objects placed on them fall to the floor when the bus is parked on a level surface. The entire interior shall be cleanable with a hose, using a liquid soap attachment. Water and soap will not normally be sprayed directly on the instrument and switch panels. All proposed interior surface and finish material and color descriptions, as well as cleaning procedures, shall be submitted to the Government for review and acceptance (see Section 3.56). All interior materials shall meet the requirements of FMVSS 302. Alternatley, interior materials shall also be in compliance with FTA Docket 90-A.

3.26.2 Interior Panels

Interior side-trim panels shall be textured anodized aluminum, melamine, or composite panels with adequate support backing. Panels shall be easily replaceable and tamper-resistant. They

shall be reinforced, as necessary, to resist vandalism and other rigors of transit bus service. Individual trim panels and parts shall be interchangeable to the extent practicable.
Unless modified by a special request, the bus interior colors shall be aesthetically pleasing and be the OEM's standard interior colors.

3.26.2.1 Modesty Panels

Sturdy divider panels constructed of durable, unpainted, corrosion-resistant material complementing the interior trim shall be provided to act as both a physical and visual barrier for seated passengers. Modesty panels, or seats and stanchions which serve as barriers, shall be located at doorways to protect passengers on adjacent seats, and along front edge of rear upper level. Design and installation of modesty panels located in front of forward facing seats shall include a handhold/grabhandle along its top edge. These dividers shall be mounted on the sidewall and shall project toward the aisle no farther than passenger knee projection in longitudinal seats or the aisle side of the transverse seats. Modesty panels shall extend no higher than 6±1.0 inches above the lower daylight opening of the side windows and those forward of transverse seats shall extend downward to a level between 1-1/2 and 1 inches above the floor. Panels forward of longitudinal seats shall extend to below the level of the seat cushion. Dividers positioned at the doorways shall provide no less than a 2-1/2-inch clearance between the modesty panel and the opened door to protect passengers from being pinched. Modesty panels installed at doorways shall be equipped with grab rails. The modesty panel and its mounting shall withstand a static force of 250 pounds applied to a four-inch by four-inch area in the center of the panel without permanent visible deformation

3.26.2.2 Operator Barrier

A barrier or bulkhead between the operator and the street-side front passenger seat shall be provided. The barrier shall minimize glare and reflections in the windshield directly in front of the barrier from interior lighting during night operation. The barrier shall extend from the floor to the ceiling and shall fit the bus side windows and wall to prevent passengers from reaching the operator or the operator's personal effects. Barrier shall be constructed of aluminum, melamine or fiberglass.
It is permissible for the operator's barrier to be integrated with the operator's equipment (radio, electrical, and storage) compartment.

3.26.2.3 Rear Panels

The rear bulkhead paneling shall be contoured to fit the ceiling, side walls, and seat backs so that any litter, such as a cigarette package or newspaper, will tend to fall to the floor or seating surface when the bus is on a level surface. Any air vents in this area shall be louvered to reduce airflow noise and to reduce the probability of trash or liter being thrown or drawn through the grille. If it is necessary to remove the panel to service components located on the rear bulkhead, the panel shall be hinged or shall be able to be removed and replaced by a 3M mechanic in 5 minutes. Grilles where access to or adjustment of equipment is required shall be heavy duty and designed to minimize damage.

3.26.2.4 Headliner

Ceiling panels shall be melamine or ABS plastic. Headlining shall be secured without loose edges and shall be supported to prevent buckling, drumming, or flexing. Where panels are in contact with metallic surfaces, headlining materials shall be treated or insulated to prevent marks due to condensation. As required to make the edges tamperproof, moldings and trim strips shall be stainless steel, aluminum, or plastic, colored to complement the ceiling material

3.26.2.5 Fastening

Interior panels shall be attached so that there are no exposed unfinished or rough edges or rough surfaces. Panels and fasteners shall not be easily removable by passengers but shall be easily replaceable when necessary.

3.26.2.6 Insulation

Any insulation material used between the inner and outer panels shall be in accordance with this specification.

3.26.2.7 Front End

The entire front end of the bus shall be sealed to prevent debris accumulation behind the dash and to prevent the operator's feet from kicking or fouling wiring and other equipment. The front end shall be free of hazardous protrusions. Paneling across the front of the bus and any trim around the operator's compartment shall be formed metal, ABS plastic or reinforced fiberglass.

Plastic dash panels shall be reinforced, as necessary, resistant to age discoloration and cracking, vandal-resistant, and replaceable. All colored, painted, and plated parts forward of the operator's barrier shall be finished with a smooth, dull matte surface that matches or coordinates with the bus interior.

3.26.2.8 Access Panels and Doors

Access for maintenance and replacement of equipment shall be provided by panels and doors that appear to be an integral part of the interior. Removal of fixtures and equipment that are unrelated to the repair task to gain access should be minimized. Where possible, access doors shall be hinged with props to hold the doors out of the mechanic's way. All door hinges shall be stainless steel.

3.27 Floor Covering

The floor covering shall be RCA Rubber Transit-Flor or Gerflor Tarabus NT, or equal.

Prior to applying rubber floor covering, all voids, fastener heads, and cracks between floor panels shall be filled, and the floor made smooth and true within 1/16 inch in any direction using a fire retardant leveling compound.

Seams shall be tightly bonded to the floor panel and have no perceivable variations in heights. At all door openings, the rubber floor covering shall connect properly and form a positively clamped, watertight seal with mating moldings/threshold plates.

Standee lines shall be marked in yellow or white at both the front vestibule area and rear door area of the bus. The standee line at the rear door area shall be a minimum of 5 inches wide.

3.28 Passenger Interior Lighting

Interior lighting shall be LED standard configuration with white lens.
When in the RUN and NITE/RUN mode, the first streetside module and complete curbside shall automatically extinguish or dim when the front is in the closed position and all interior modules shall light when the front or rear doors are opened.

Entry Area Lighting

The bus shall be equipped with lights to illuminate the doorways and adjacent platform for the purpose of safe boarding and disembarking of passengers.

3.29 Passenger Seats

3.29.1 Passenger Seats General

There shall be a minimum total of 34 passenger seats, including fold-down seats to accommodate wheelchairs. The seating configuration shall be perimeter type and shall not exceed seven (7) forward facing seats. (Some options may reduce seating capacity) While it is anticipated that these buses will not require passenger seat belts, if after the order is placed, the manufacturer determines that FMVSS compliance requires seat belts, they will be added at additional cost negotiated between the Government and the vendor.

The seats shall include stainless steel frames with padded inserts (meeting FMVSS 302 and FTA Docket 90-A). The successful offeror shall furnish a seating layout drawing which will be approved by the Government.

Seat upholstery design and color shall be a stain hiding blend and shall be the OEM's standard upholstery complimenting the interior color scheme.

As applicable, hip-to nee room measured from the front of one seat back horizontally across the highest part of the seat to the seat or panel immediately in front shall be no less than 26 inches. At all seating positions in paired transverse seats immediately behind other seating positions, hip-to-knee room shall be no less than 26 inches. Seat height, except for rear bench seat, shall be 17 inches. The rear bench seating height shall not exceed 21 inches.

Transverse mounted seats shall, wherever possible, be fully cantilever mounted.

3.29.2 Wheelchair Accommodation

Two forward-facing wheelchair securement locations and a total of two securements, as close to the wheelchair loading system as practical, shall be provided for each bus. Each wheelchair accommodation shall provide parking space and a secure tie down (compliant with ADA requirements) for two passengers in wheelchairs.

3.29.3 Maneuvering Room

Maneuvering room inside the bus shall accommodate ease of travel for a passenger in a wheelchair from the loading device through the bus to the designated parking area and back out per ADA requirements.

Additional equipment, including passenger restraint seat belts, shoulder harnesses and wheelchair securement devices shall be provided for each wheelchair passenger. Passenger restrain seat belts shall be provided to accommodate passengers in electrically-powered wheelchairs. All belt assemblies must stow up and out of the way when not in use.

3.29.4 Wheelchair Securing Devices

Two wheelchair securing devices shall be provided. The installation of wheelchair securement device shall not require excessive bending or twisting by the operator. The jump seat release mechanism shall be easily operable. The latching mechanisms for wheelchair wheels shall not interfere with battery-operated wheelchairs.

3.29.5 Seat Belts

Built-in passenger restraint seat belts shall be provided in each wheelchair parking area. Seatbelts shall be easily accessible for wheelchair users. On the curbside securement area a shoulder belt should be attached to the sidewall of the bus. A belt-type securement system and shoulder strap seat belt shall be included. The latching mechanism and retracted belts shall be readily visible when seats are folded down. Wheelchair area accommodations shall comply with the latest ADA laws and federal safety requirements.

3.30 Passenger Assists

Stanchions and handrails shall be provided to aid passengers when boarding, moving through the vehicles, and standing while onboard the vehicles.

Stanchions shall not be placed where they will obstruct the passenger flow through the bus vehicle doors. Stanchions shall be designed to adequately withstand the loads that may be applied to them (up to 300 lbs. at midpoint in any direction) without permanent deformation. Stanchions and handrails shall be brushed stainless steel or approved equal. Stanchions and handrails shall comply with ADA regulations and shall be between 1.25 inches and 1.5 inches in diameter unless otherwise approved.

3.31 Wheelchair Loading System

3.31.1 General

A loading system shall accommodate passengers on wheelchairs or using crutches, canes, walkers, or persons with difficulty using steps for ingress and egress from or to the street level or curb quickly, safely, and comfortably. The wheelchair loading system shall conform to all applicable ADA requirements.

3.31.2 Wheelchair Ramp

A front door located, automatically-controlled, ramp system, ADA compliant, meeting requirements defined in 49 CFR Part 38, shall provide ingress and egress quickly, safely, and comfortably, both in forward and rearward directions, for a passenger in a wheelchair from a level street or curb. The ramp shall be of a simple hinged, flip-out type design. The deployed ramp shall be 30 inches wide (minimum) and 44 inches long (minimum).

When the system is not in use, the passageway shall appear normal. In the stored position of the ramp, no tripping hazards shall be presented and any resulting gaps shall be minimized. The controls shall be simple to operate with no complex phasing operations required, and the loading system operation shall be under the surveillance and complete control of the operator. The bus shall be prevented from moving during the loading or unloading cycle by a throttle and brake interlock system. The wheelchair loading system shall not present a hazard, nor inconvenience any passenger. The loading system shall be inhibited from retracting or folding when a passenger is on the ramp/platform. A passenger departing or boarding via the ramp shall be able to easily obtain support by grasping the passenger assist located on the doors or other assists provided for this purpose. The platform shall be designed to protect the ramp from damage and persons on the sidewalk from injury during the extension/retraction or lowering/raising phases of operation. The loading platform shall be covered with a replaceable or renewable, nonskid material and shall be fitted with devices to prevent the wheelchair from rolling off the sides during loading or unloading. A manual override system shall permit unloading a wheelchair and storing the device in the event of a primary power failure.

3.32 Operator Area

The objective of the design of the operator's area is to provide an environment for the operator to operate the bus safely and efficiently for long periods of time without injury and with minimal fatigue. The operator's area shall therefore be designed to minimize operator glare. The use of polished metal and light-colored surfaces within and adjacent to the operator’s area shall be avoided. Areas that are visible from the outside of the bus in the vicinity of the dash panel and cowl shall be configured to preclude use as storage of items.
The operator's area shall comply with the following SAE recommended practices, including:

- SAE J287, Driver Hand Control Reach;
- SAE J941, Motor Vehicle Driver Eye Range;
- SAE J1050, Driver's Field of View;
- SAE J1052, Motor Vehicle Driver and Passenger Head Position;
- SAE J1516, Accommodation Tool Reference Point;
- SAE J1522, Truck Driver Stomach Position.

3.32.1 Operator Controls/Instrumentation

All switches and controls necessary for the safe and efficient operation of the bus vehicle shall be conveniently located in the operator's area, grouped according to function. Controls shall be located so that boarding passengers may not easily tamper with control settings. Similarly, all switches and controls shall not pose a hazard to the Operator. Guards shall be provided on switches as necessary to prevent inadvertent activation. The nomenclature of each indicator shall be readable when either energized or de-energized. All controls/instruments shall be placed and protected to prevent damage by wind or rain when any bus vehicle windows/doors are open, whether the bus vehicle is moving or is stationary.

3.32.2 Controls

As located in ergonomically convenient locations for the operator, the following identified switches and lights shall be visibly accessible and include (at a minimum):

- Master run switch;
- Start button or switch;
- Kneel switch;
- Interior lighting switch;
- Instrument panel lighting intensity control (solid state);
- Operator's area light switch;
- Hazard warning switch;
- Climate controls;
- Defroster control;
- Operator's heater control;
- Parking emergency brake control (brake activation indicated);
- Front door 'dump' valve;
- PA system controls;
- High beam;
- Exit door open or unlocked;

- Horn button in steering wheel hub;
- Headlight dimmer switch (foot controlled);
- Fire suppression;
- Wheelchair ramp switches and ramp control switch;
- Operator’s fan;
- Engine 'Stop' override;
- Front dome light.

These controls shall be identifiable by shape, touch, and permanent markings. All switches shall be heavy-duty push button, rocker, rotary (spring return to center) or toggle switches. Requests for other types of switches or controls shall be submitted to the Government for review and approval. Door control, kneel control, windshield wiper/washer controls, and run switch controls shall be the most convenient for the operator to access. The hazard-warning switch must be longer or physically different thereby allowing the operator to quickly find its location and activate it merely by feeling for the switch. Doors shall be operated by a single control, conveniently located and operable in a horizontal plane by the operator's left hand. The setting of this control shall be easily determined by position and touch. Turn signal controls shall be floor-mounted, foot-controlled, water resistant, heavy-duty, momentary contact switches.
Switches, controls, and instruments shall be dust- and water-resistant consistent with the specified bus interior cleaning practice.

3.32.3 Instruments

The speedometer, certain indicator lights, and air pressure gauge(s) with two needles shall be a minimum of two (2) inches in diameter and shall be located on the front cowl immediately ahead of the steering wheel. All gauges shall have a verifiable (by Government) operating accuracy. Illumination of the instruments shall be simultaneous with the marker lamps. Glare or reflection in the windshield, side window, or front door windows from the instruments, indicators, or other controls shall be minimized. Instruments and indicators shall be easily readable in direct sunlight. The use of backlit instrumentation and gauges shall be maximized.

The instrument panel shall include an electric analog speedometer or digital display with a maximum possible indicating speed beyond 65 mph and calibrated in maximum increments of 5 mph.

The instrument panel shall also include an air pressure gauge with indicators for primary and secondary air tanks, voltmeters for 24 volt systems, engine oil pressure gauge, and engine coolant temperature gauge. The instrument panel and wiring shall be easily accessible for service from the operator's seat or top of the panel. Wiring shall have sufficient length and be routed to permit service without stretching or chafing the wires. An audible alarm shall be furnished to alert the operator of excess engine coolant temperature and low oil pressure.

3.32.4 Indicators/On-Board Diagnostics

Critical systems or components shall be monitored by a built-in diagnostic system with visible and audible indicators. The diagnostic indicator lamp panels shall be located in clear sight of the operator. The intensity of indicator lamps shall permit easy determination of on/off status in bright sunlight and shall not cause a distraction or visibility problem at night. All indicators shall have a method of momentarily testing the operation of the lamp. The successful offeror shall submit the design, installation, and operating details of the proposed diagnostic system indicator and on-board diagnostics display(s) to the Government for review and acceptance (see Section 3.56).
Whenever possible, sensors shall be of the closed-circuit type so that failure of the circuit or sensor shall activate the malfunction indicator. Audible alarms shall be loud enough for the operator to be aware of its operation and be inclined to discontinue operation of the bus. Malfunction and other indicators listed in Table 2-Onboard Diagnostic Indicators shall be provided for the bus:

Table 2: Onboard Diagnostic Indicators

Visual Indicator	**Audible Alarm**	**Condition or Malfunction**
ABS	None	ABS System Malfunction
Check Engine	None	Engine Electronic Control Unit detects a malfunction
Stop Engine	None	Engine interlocks
Check Transmission	None	Transmission Electronic Control Unit detects a malfunction
Charging System	None	Alternator not charging
Fire	Bell or Buzzer	Over-temperature/fire detection condition in engine.
Reserved		
Reserved		
Reserved		
Reserved		
Reserved		
Hazard Warning	Audible Clicks	Warning signal to other operators (may be common with turn indicators)
High Headlamp	None	High headlamp is on

Hot Engine	Buzzer	Excessive engine coolant temperature
Reserved		
Kneeling	Buzzer	Kneeling system activated
Low Fuel	None	Low Fuel Pressure
Low Coolant	Buzzer	Insufficient engine coolant level
Low Charge	None	Low charge conditions
Low Air	Buzzer	Insufficient air pressure in either primary or secondary reservoirs
Low Oil Pressure	Buzzer	Insufficient engine oil pressure
Mobility Aid Passenger Exit Signal	None	Mobility aid passengers want to disembark
Parking Brake Applied	None	Parking brake is applied
Service Brake Applied	None	Service brake is applied (may be common with parking brake indicator)
Rear Doors Open	Buzzer	Rear doors are opened
Reserved		
Right and Left Turn	Audible Clicks	Indication of left or right turn
Wheelchair Ramp	Beeper	Wheelchair ramp is not stowed and disabled

3.32.5 Door Controls

Controls for the front and rear doors shall be a single, 4 or 5-position master door switch, conveniently located and operable in a horizontal plane by the operator's left hand. The setting of this control shall be easily determined by position and touch. Other concepts may be proposed for Government consideration. The master door switch shall be a single, 4 or 5-position, control with the following settings provided:

Switch/Lever Position	Front Door	Rear Door
1st Position Rearward	Closed	Opened
Center Position	Closed	Closed - Locked
1st Position Forward	Opened	Closed – Locked
2nd Position Forward or Rearward	Opened	Opened

3.32.5.1 Door Operations

The front bus vehicle doors are to be 'driver open' and 'driver closed' commanded. Power for opening and closing the front doors shall be done either pneumatically or electrically. The door shafts must be designed to mitigate operational problems (such as 'jumping' out of their bearings). The rear door is also to be 'driver open' and 'driver closed' commanded. Opening the rear door shall be conducted either pneumatically or electrically with a single cylinder, closed with a spring or air cylinder, and locked with an electro-mechanical lock.

No more than a 10 pound force shall be imposed on a 1 square inch area of any passenger struck by a closing rear door. Sensitive edges are to be verified with a 1.5-inch obstruction that simulates a small child's arm. A maximum force of 25 pounds shall be required for a passenger to free himself after having either door close upon him/her. The opening and closing of all doors should take no more than 5 seconds.

Door mechanisms shall be concealed from passengers but shall be easily accessible for servicing. All elements of the door and actuator systems shall operate trouble-free for the life of the vehicle.

For air-powered doors, a valve, located at the top of the left side panel and accessible through the operator's window, shall shut off the air power to, and dump the air power from, the front door mechanism to permit manual operation of the front door with the bus shut down. Operation and location of the door dump valve shall be accepted by the Government during preliminary design review activities (see Section 3.56).

3.32.5.2 Interlock

An accelerator interlock system shall be furnished that locks the accelerator in the closed position and a braking interlock shall engage the rear axle spring brake system in accordance with the following door and door control lever positions:

Door Lever Position	Front Door	Rear Door	Interlocks
1st Position Rearward	Closed	Opened	Actuated
Center Position	Closed	Closed – Locked	Not Actuated
1st Position Forward	Opened	Closed – Locked	Not Actuated
2nd Position Forward or Rearward	Opened	Opened	Actuated

3.32.5.3 Interlock Override Switch

An interlock override switch shall be integrated to release and deactivate all interlocks. The override switch shall be such as to prevent unintentional activation. Access to the interlock override switch shall be accepted by the Government during preliminary design review activities (see Section 3.56).

3.32.6 Steering Wheel and Horn Button

The steering wheel shall last the life of the bus and shall be constructed of a hard, smooth material impervious to cleaning fluids, and body acids. The steering wheel shall be of an effective diameter and shall be shaped for firm grip with comfort for long periods of time. The steering wheel spokes or rim shall not obstruct the operator's vision of the instruments on the panel when the steering wheel is in the straight-ahead position. The steering column shall be capable of horizontal and vertical adjustment from the operator seat. Clearance requirements shall be met at all positions with adjustment ranges.
Dual electric horns shall be provided. They shall be mounted to prevent the entry of water and dirt into horn trumpets, the horns shall sound high and low notes that are clearly audible over 75 dBA noise at a distance of 300 feet. The horn button shall be located in the steering wheel hub and shall be protected from debris accumulation.

3.32.7 Accelerator and Brake Pedals

Accelerator and brake pedals shall be designed for comfortable ankle motion and shall meet the requirements of SAE J1516. Foot surfaces of the pedals shall be faced with wear-resistant, non-skid, replaceable material. Force to activate the brake pedal control shall be an essentially linear function of the bus deceleration rate and shall not exceed 70 pounds at 7 inches above the heel point of the pedal to achieve maximum braking. The heel point is the location of the operator's heel when foot is rested flat on the pedal and the heel is touching the floor or heel pad of the pedal.

3.32.8 Master Run Switch

Controls for engine operation shall be closely grouped within the operator's area. These controls include a separate master run switch and start switch or button. The master run switch shall be a four-position rotary switch (or equivalent) conveniently located to the operator's left with the following functions:

STOP ENGINE - All electrical systems shall be turned off by the PLC after a 30 minute delay, except hazard lights, radio, systems which must remain energized, and farebox if equipped

NIGHT PARK - All electrical systems off, except those listed in OFF and power to destination signs, interior lights, radio and marker lights.

DAY RUN - All electrical systems and engine on, except the headlights, parking lights and marker lights. The starter shall be inoperative when the engine is running.

NIGHT RUN - All electrical systems and engine on. The starter shall be inoperative when the engine is running.

3.32.9 Turn Signal

Turn signal controls shall be foot controlled, waterproof, heavy-duty momentary contact switches, and floor-mounted on a platform inclined to minimize confusion among the left, right, and high-beam switches.

3.32.10 Operator’s Area Lighting

The operator’s area shall have a light to provide general illumination and it shall illuminate the half of the steering wheel nearest the operator to a level of 10-15 foot candles. This light shall be operator controlled by a toggle switch located on the operator’s control panel.

3.33 Operator’s Seat

The operator’s seat shall be an air suspension USSC 9100 ALX or Recaro Ergo Metro seat with flip-up right arm rest and three point seat belt. Seat upholstery shall be black with transit cloth seat upholstery.

3.34 Operator’s Vent and Defroster

The bus vehicle interior climate control system shall deliver at least 500 cubic feet per minute of air to the operator's area when operating in the ventilating mode. Adjustable nozzles shall permit variable distribution or shut down of the air flow. Air flow in the heating mode shall be reduced proportionally to the reduction of air flow into the passenger area.

The windshield defroster unit shall meet the requirements of SAE Recommended Practice J382 and shall have the capability of diverting heated air to the operator's feet and legs. The defroster or interior climate control system shall maintain visibility through the operator's side window. A draining tube shall be installed if the defroster or operator's heater is to be in an enclosed box.

3.35 Auxiliary Windshield De-Mister Fans

Two auxiliary, two-speed fans shall be at the front windshield area. If mounted on the dash, fan bases shall be mounted to a tapping plate that is installed to reinforce the dashboard. These plates shall be attached by welding or with machine screws. No rivets are allowed. Each fan shall be equipped with a "HIGH-OFF-LOW" toggle switch located convenient to the driver to selectively control the operation of the fan.

3.36 Interior Mirrors

Mirrors shall be provided for the operator to observe passengers throughout the bus without leaving his seat and without shoulder movement. With a full standee-load, including standees in the vestibule, the operator shall be able to observe passengers in the front and rear door area, anywhere in the aisle, and in the rear seats. Interior mirrors shall not be in the line of sight to the right outside mirror. Mountings shall be sturdy to resist flexing, vibration, and vandalism.

Interior observation shall be accomplished by a swivel-mounted mirror no less than 4 inches by 16 inches attached above and to the right of the operator's head. The rear exit shall be observed by the use of two mirrors. One shall be a 6-inch round mirror installed at the upper right corner near the windshield header. The second shall be a 12-inch convex mirror installed at the rear exit. Mirrors mounted near the entrance or exit shall not be an obstruction to passengers and shall provide maximum visibility to the operator.

3.37 Exterior Mirrors

The bus shall be equipped with a corrosion-resistant exterior rearview mirror on each side of the bus. Mirrors shall permit the operator to view the roadway along both sides of the bus, including the rear wheels. Each mirror shall be heated and shall feature motorized adjustment (being remote controlled in an ergonomically efficient manner relative to a seated operator). Mirrors shall be firmly attached to the bus to prevent vibration and loss of adjustment, but not so firmly attached that a bus or its structure is damaged if the mirror is struck, and shall retract or fold sufficiently to allow bus washing operations. Mounting arms shall not protrude beyond the outside mirror edge.

3.38 Public Address System

A public address system that complies with the ADA requirements of 49 CFR, Part 38.35 and enables the operator to address passengers either inside or outside the bus shall be provided.

The system shall be comprised of one (1) audio amplifier, one (1) minimum 30 inch gooseneck mounted microphone with on/off switch, loudspeakers as necessary, one (1) Inside/Outside selector switch, and cable, connectors, and hardware as necessary to install a working system as described herein.

Inside speakers shall broadcast, in a clear tone, announcements that are clearly perceived from all seat positions at approximately the same volume level. The loudspeakers shall be located inside the bus and distributed in a fashion so as to provide a nominal level at each seat with no more than 3 dbA variation seat-to-seat. The loudspeakers selected for use inside the bus shall meet the following minimum requirements:

A speaker shall be provided so announcements can be clearly heard by passengers standing outside each bus near the front door. One or more loudspeakers shall be provided and installed where necessary to provide announcements to passengers outside the bus and within 25' of the bus doors. These loudspeakers shall be impervious to all types of weather as well as bus wash operations. Loudspeakers shall be selected and installed so that they do not obstruct the view of any seated passenger or the driver. The loudspeakers shall not be selected, located, or installed in such manner that they may cause harm or injury to any person walking, standing, or sitting near any loudspeaker in the course of normal use and operation.

The public address system shall be self muting when not in use and powered from a source which does not have adverse electrical noise. The contractor shall employ measures as necessary to defeat any noise intrusion into the sound system such as pops, buzzes, clicks, hum, hiss, and ignition noise as attributed but not limited to power source, alternator, generator, or lighting noise without alteration to the sound system performance.

The microphone shall be vandal resistant and mounted on a heavy-duty flexible gooseneck, minimum of 24 inches long, and shall allow the operator to comfortably speak into it without holding it. The gooseneck mounting flange shall be located at the contractor's standard mounting location.

3.39 Reserved

3.40 Storage Locker

The contractor shall furnish and install a minimum of one operator's storage locker. The locker(s) shall be furnished with finger operated latches-no keyed locks.

3.41 Operator's Platform

The operator's platform shall be finished with no sharp edges and shall not interfere with or impede wheel chairs or other mobility aids. The floor in the operator's area shall be easily cleaned and shall be arranged to prevent debris accumulation.

3.42 Front Curbside Wheelwell Cover and Package Rack

The front curbside wheelwell cover shall be a “square top” wheelwell cover furnished with a large stainless steel package rack.

3.43 Heating, Ventilation, and Air Conditioning

3.43.1 Configuration

The HVAC unit shall be a rear mounted Thermo King T series, or roof mounted Thermo King RLF series, or Thermo King Athenia AMII series.
The Thermo King system shall be furnished with the following:

- S391 Rotary Screw Compressor
- Brushless evaporator and condenser fan motors
- R-407C chlorine free, non-ozone depleting refrigerant
- IntelligAIRE® III Total Climate Control system (includes RS-232 serial interface)

Manually controlled shutoff valves in the refrigerant lines located on either side of all circulating/booster pumps shall be implemented to facilitate ease of service. Self-sealing couplings utilizing O-rings seals, or shut off valves, shall be used to break and seal the refrigerant lines during major components removal.

The condenser shall be located to efficiently transfer heat to the atmosphere without ingesting air warmed by other components and discharging air to other components. The condenser shall also be obstruction free from wheel splash, road dirt, or debris.

All HVAC components located within 6 inches of floor level shall be constructed to resist corrosion and damage.

3.43.2 Performance and Capacity

With the bus operating at its design operating profile and door opening/closing cycle, and at 150 % of seated load, the HVAC system shall maintain an average passenger compartment temperature between 65 degrees and 80 degrees Fahrenheit, and a relative humidity of 50% or less.

When the bus is operated in outside ambient temperatures of 95 degrees to 115 degrees Fahrenheit, the interior temperature of the bus shall be permitted to rise 1 degree for each degree of exterior temperature in excess of 95 degrees Fahrenheit.

At an outside ambient temperature range of -10º to +110° F, the interior temperature of the bus shall not fall below 55º Fahrenheit, while operating on the Design Operating Profile. If required to meet these requirements, an auxiliary coolant heater shall be furnished.

The air conditioning portion of the HVAC system shall be capable of reducing the passenger compartment temperature from 110º to 90º F in less than 20 minutes after engine start-up. Engine temperature shall be within the normal operating range at the time of start-up of the cool-down test and the engine speed shall be limited to fast idle that may be activated by an operator-controlled device. During the cool-down period the refrigerant pressure shall not exceed safe high-side pressures and the condenser discharge air temperature, measured 6 inches from the surface of the coil, shall be less than 45º F above the condenser inlet air temperature. The appropriate solar load as recommended in the APTA "Recommended Instrumentation and Performance Testing for Transit Bus Air Conditioning System," representing 4 P.M. on August 21, shall be used. There shall be no passengers on board, and the doors and windows shall be closed.

The pull up requirements for the heating system shall be in accordance with Section 9 of APTA's "Recommended Instrumentation and Performance Testing for Transit Bus Air Conditioning." With an ambient temperature of -15° F, and the bus cold soaked at that temperature, the bus heating system shall warm the center passenger compartment to an average temperature of 70°±2°F within 70 minutes. Option AH (Auxiliary Heater) may be utilized if needed.

Testing shall be conducted in accordance to the APTA Recommended Instrumentation and Performance Testing for Transit Bus Air Conditioning System. Temperature measurements shall be made in accordance to this document with the following modifications:

The three primary locations used for temperature probes are (1) 6 inches aft of front wheel housing, (2) centered between the two axles and (3) 6 inches aft of rear wheel housing. At each primary location, the nine (9) temperature sensing devices shall be (A) 72 inches above floor level, (B) 6 inches above top surface of seat cushion and (C) 6 inches above floor.

The recommended locations of temperature probes are only guidelines and may require slight modifications to address actual bus design. Care must be taken to avoid placement of sensing devices in immediate path of air duct outlet. In general, the locations are intended to accurately represent the interior passenger area.

3.43.3 Controls

The HVAC system, excluding driver's heater and defroster, shall be centrally controlled with an advanced electronic/diagnostic control system having provisions for extracting data via a portable testing unit.

The temperature control set-point for the system shall be 70 degrees Fahrenheit. The entire HVAC system shall be operated fully automatically to control the interior passenger compartment average temperature to within 2 degrees Fahrenheit of the aforementioned temperature control set-point. In the event of compressor or condenser fan failures, the control system shall automatically switch to ventilating mode.

Temperatures measured from a height of 6 inches below the ceiling shall be within ±5 degrees Fahrenheit of the average temperature at the top of the seat surfaces. Temperatures measured more than 3 inches above the floor shall be within ± 10 degrees Fahrenheit of the average temperature at the top of the seat surfaces. The interior temperatures measured from front to rear of the bus shall not vary more than ± 5 degrees Fahrenheit from the average when in heating mode.

Vents shall be provided to supply conditioned air into the driver's area. The vents and fans leading to the operator's area shall be controlled by the operator. Further, the operator shall have full control over the driver's heater and defroster, and the temperature in his/her area shall be adjustable through air distribution and fans.

3.43.4 Air Flow

Ventilation shall be accomplished by the blower fans that are integral within the HVAC system.

The cooling mode of the HVAC control system shall recirculate air within the bus at or near the ceiling height at a minimum rate of 25 cubic feet per minute (cfm) per passenger based upon the standard configuration bus carrying a number of passengers equal to 150 percent of the seated/standing load. Air flow shall be evenly distributed throughout the bus with air velocity not exceeding 100 feet per minute on any passenger. The ventilating mode shall provide air at a minimum flow rate of 20 cfm per passenger.

Air flow may be reduced to 15 cfm per passenger (150 percent of seated load) when operating in the heating mode. The fans shall not activate until the heating element has warmed sufficiently to assure 70 degrees Fahrenheit air outlet temperature. The heating air outlet temperature shall not exceed 120 degrees Fahrenheit under any normal operating conditions.

Outside air flow may be cut off during initial warm-up, provided no manual manipulation is required.

In addition, the HVAC system shall deliver at least 100 cfm of air to the operator’s area when operating in the ventilating and cooling modes. Adjustable nozzles shall permit variable distribution or shutdown of the airflow. Airflow in the heating mode shall be reduced proportionally to the reduction of airflow into the passenger area.

The windshield defroster unit shall meet the requirements of SAE Standard J382 and be able to divert heated air to the operator’s feet and legs. The system shall also maintain visibility through the operator’s side window.

3.43.5 Reserved

3.43.6 Emergency Escape Hatches

Two combination roof ventilator/escape hatches, Transpec® Worldwide model 1000, shall be provided in the roof of each bus vehicle-one approximately over the rear axle and one approximately forward of the front axle. The ventilator/hatch shall comply with the requirements of FMVSS 217. The ventilator/escape hatches shall cover an opening area no less than 425 square inches and shall be capable of being positioned as a scoop with either the leading or trailing edge open or with all four edges raised simultaneously.

3.43.7 Air Intakes

Outside openings for air intake shall be in a location designed to ensure cleanliness of air entering the HVAC system, particularly with respect to exhaust emissions from the bus, adjacent traffic, and airborne dust generated by the rear wheels. All intake openings shall be baffled to prevent entry of water.

In addition, outside air shall be filtered before discharge into the passenger compartment. The filter shall meet ASHRAE requirement for 5 percent or better atmospheric dust spot efficiency and a minimum dust holding capacity of 120 grams per 1000-cfm cell. More efficient air filtration may be provided to maintain efficient heater and/or evaporator operation. Air filters shall be easily removed for service. Moisture drains from air intake openings shall be located to prevent clogging from road dirt.

3.43.8 Entrance/Exit Area Heating

Heat shall be supplied to the entrance and exit floor areas of the proposed bus vehicle to prevent accumulation of snow, ice, or slush with bus operating under design operating profile and corresponding door opening cycle.

3.44 Signage

All electrical materials, components and installation methods used in the fabrication of the entire communication/signage system, including wire, cable, printed circuit boards, antennas, and circuit protection devices shall conform to this Section.

3.44.1 Destination Signs

3.44.2 General

The contractor shall provide an ADA-compliant visible passenger information system providing announcements outside of bus vehicle. Control of displays shall be integrated as a dedicated subsystem unit into the overall communication system. All messages shall be easily updated by utilizing a plug-in hand-held device. Two such devices shall be furnished per bus.
The display shall be electronic, alphanumeric, and completely self-contained. All passenger information shall be ADA compliant and shall consist of a minimum of 15 characters and a minimum of 60mm (2.4 in.) high. Information displays shall be pitched slightly downward to allow maximum visibility to passengers.

The displays shall be readable in direct sunlight or complete darkness and all degrees of light in between. The horizontal and vertical readable viewing angles shall be a minimum of 170 degrees. The display shall have a minimum contrast ratio, in daytime, of 35:1. The displays shall not require any external mask between pixels and shall not require any framing or support structure between characters which would give the sign a discontinuous appearance.

Power to the sign shall be controlled by the Master Run Switch. The signs shall operate in all positions of this switch except "OFF". Sign lighting shall operate in the "RUN" and park positions. The sign shall be internally protected against voltage transients and RFI interference to ensure proper operation in the transit operating environment.

3.44.3 Front and Side Destination Sign

One LED or LCD exterior display shall be provided per vehicle on the front above the windshield. In addition, an identical display shall be provided on the side of the vehicle adjacent to the entrance door to allow for passenger viewing prior entering the vehicle. The front destination sigh shall be furnished with a defroster grid.

Design and operating parameters shall be submitted by the successful contractor to be reviewed and approved by the Government (see Section 3.56).

3.44.4 Other Numbering and Signage

Signs shall be durable and fade, chip, and peel resistant - they may be painted signs, decals, or pressure-sensitive applied. All decals shall be sealed with clear waterproof sealant around all exposed edges (as required). Signs shall be provided in compliance with applicable ADA requirements as defined in 49 CFR subpart B, 38.27.

3.45 Passenger Information Holder

Provisions shall be made at the rear of the operator's barrier for a frame to retain information posted by Government, such as routes and schedules. The retainers may be concave and shall support the media without adhesives.

In addition, provisions shall be furnished for interior bus cards, which permit the Government to install ad-style displays running up both sides of the interior of the bus. These display holders shall support the display media without adhesive.

3.46 Electrical

3.46.1 Power requirements

The vehicle master run switch shall not carry circuit operating loads but shall control load carrying relays or magnetic switches.

3.46.2 Wiring General Requirements

Wire sizes, insulation requirements, materials, shielding methods, and identification of wire and cable used for primary, auxiliary, control, and communications applications shall be based on the current carrying capacity, voltage drop, mechanical strength, temperature, and flexibility requirements of SAE J1292.
All wire and cable insulation shall meet the flame and smoke test requirements of this specification and be substantially free of halogens. The contractor shall mark each wire every foot by wire type and each wire end with a function code using a scheme.

Braided copper wire shall be used in all ground strap applications. Flexible stranded copper wire is acceptable in other applications. All conductors of multi-conductor cables shall be terminated. The shields on multi-conductor low voltage control cables shall be terminated on one end only to minimize the effects of ground-loops.

3.46.3 Wiring and Terminals

All lamp sockets shall be of two-wire design with disconnects to eliminate corrosion or ground problems. To facilitate servicing, all lamp wires shall have leaders of at least 6 inches.

All exposed wiring between major electrical components and terminations shall have double electrical insulation, shall be waterproof, and shall conform to specification requirements of SAE Recommended practice J1127 and J1128. The requirement for double insulation shall be met by sheathing all insulated wires and harnesses with non-conductive rigid or flexible conduit.
Except as interrupted by the master battery disconnect switch, battery and starter wiring shall be grouped and numbered and/or color coded with connections secured by bolted terminals and shall conform to specification requirements of SAE Standard J1127-Type SGT, SGX, or GXL and SAE recommended practice J541. Wiring shall be color coded and/or coded with a configuration code to determine circuitry. Wiring numbers shall be marked every 6 inches. Insulation shall permit ease of replacement. All wiring harnesses over 5 feet long and containing at least 5 wires shall include 10 percent excess wires for spares.

Wiring in exposed areas shall be double insulated. Double insulation shall be maintained as close to the terminals as possible. The requirement for double insulation shall be met by sheathing all wires and harnesses with non-conductive rigid or flexible conduit. Strain-relief fittings shall be provided at points where wiring enters all electrical components. Grommets of elastomeric material shall be provided at points where wiring penetrates metal structures outside of electrical enclosures and all other apertures in the vehicle. Any clamps used throughout the electrical system shall be stainless steel and of aircraft-type quality and shall be "dipped" and shall be spaced no greater than 24 inches on center. Individual tie wraps may be used to bundle wires and cable, but shall not be used to support the bundles; however, tie wrap type clamps are approved for use where it is the OEM commercial practice to utilize such clamps for support. Wiring supports shall be non-conductive. Precautions shall be taken to avoid damage from heat, water, solvents, or chafing. All wiring shall be kept at least 15cm (6 inches) away from the exhaust system and engine turbocharger, unless protected by appropriately rated reflective shields, heat resistant tubing, or high temperature loom. Wiring length shall allow for replacement of end terminals twice without pulling, stretching, or replacing the wire. Except for large wires such as battery cables, terminals shall be crimped to the wiring and may be soldered only if the wire is not stiffened above the terminal and no flux residue remains on the terminal. Terminals shall be corrosion-resistant, full-ring type or interlocking lugs with insulating ferrules. "

Connectors shall be common, and self-aligning. Some suitable manufacturers include Delphi/Packard, Amphenol, Amp, Molex and Deutsch.

3.46.4 Junction Boxes and Fittings

Boxes, covers, and fittings may be constructed of aluminum, fiberglass or galvanized ferrous metals. Screws and other hardware shall be made of stainless steel.

3.46.5 Cleating

Any low voltage cables and wiring not installed in conduits shall be cleated by split block cleats of molded neoprene rubber or an approved equivalent. A nonflammable insulating material with a durometer reading of 50 to 60 shall be used for cleating. The holes in the cleat shall be sized for the individual wires and cables.

Each cleat shall spread the bolt clamping force over the entire length of the cleat. Bolts shall have lock nuts of approved design. Cable and wiring using cleating shall be supported to the bus vehicle body at least every 24 inches, except cleating supporting conduit shall be spaced as required to prevent conduit sag and movement.

3.46.6 Splicing and Taping

Splicing and taping shall not be allowed unless approved by the Government

3.46.7 Circuit Protection

All main and branch circuits, except battery-to-starting motor and battery-to- alternator circuits, shall be protected by circuit breakers or fuses sized to the requirements of the load. Electronic circuit protection or controls for the cranking motor shall be provided to prevent overheating. The circuit breakers or fuses shall be easily accessible for authorized personnel. Fuses shall be used only where circuit breakers are not practicable. Any manually re-settable circuit breakers shall provide visible indication of open circuits. Circuit breakers shall meet the requirements of SAE J553 or SAE J1625, or UL489/UL1077.

Circuit breakers or fuses shall be sized to a maximum of 15 percent larger than the total circuit load current. The current rating for the wire used for each circuit must exceed the size of the circuit protection being used.

Handles shall indicate ON, OFF, and TRIPPED positions, or alternatively, only ON, OFF if there is no "TRIPPED" position . Fail-safe automatic reset circuit breakers shall be employed when it is necessary to quickly re-establish circuit continuity when that portion of the wiring has been subjected to an overload condition. Circuit breakers shall be molded-case type, single- or multi-pole, with frame size suitable for continuous current and interrupting duty. Each pole shall be equipped with a trip mechanism consisting of an inverse time element for overload protection and an instantaneous magnetic element for short circuit protection. Each pole shall be equipped with adequate means of arc extinction to prevent flashover. Multi-pole breakers shall operate contacts simultaneously.

Breaker current rating shall be clearly visible after installation.

3.46.8 Grounding

Dedicated insulated wires and/or copper braided straps shall be used for all electrical equipment, except where it can be demonstrated that dedicated grounds are not feasible or practicable. One ground plane may be the bus vehicle body and framing. Ground conductors and terminals can be clustered and shall be accessible for servicing. Grounding terminals shall be attached to ground conductors in conformance with SAE J163. No more than four (4) terminals shall be connected to a ground bolt. Grounds shall not be carried through water piping, hinges, bolted joints (except those specially designed as electrical connectors), or power plant mountings. Electrical equipment shall not be located in an environment that will reduce the performance or shorten the life of the component or electrical system. Major wiring harnesses shall not be located under the bus floor, and under floor wiring shall be eliminated to the extent practicable. Wiring and electrical equipment necessarily located under each bus shall be insulated from water, heat, corrosion, and mechanical damage, and shall be contained in sealed rigid or flexible conduit.

3.46.9 Electrical Components

Grounds, using dedicated insulated wires and/or copper braided straps, shall be connected directly to the vehicle chassis, not through hinges or panels. Ground conductors and terminals can be clustered and shall be accessible for servicing. Grounding terminals shall be attached to ground conductors in conformance with SAE J163. Grounds shall not be carried through water piping, hinges, bolted joints (except those specially designed as electrical connectors), or power plant mountings. Electrical equipment shall not be located in an environment that will reduce the performance or shorten the life of the component or electrical system. Major wiring harnesses shall not be located under the bus floor, and underfloor wiring shall be eliminated to the extent practicable. Wiring and electrical equipment necessarily located under each bus shall be insulated from water, heat, corrosion, and mechanical damage, and shall be contained in sealed rigid or flexible conduit.

3.46.10 Batteries

Two lead-acid batteries, meeting the following requirements, shall be provided for 12/24 volt service:

- Interstate No. U8D-1400BUS, or equal, 12 Volt, 430 RC min. @ 25amp @ 80 deg. F.

They shall bear an initial warranty date no earlier than 60 days prior to bus manufacture. Positive and negative terminals shall have different size studs, or the battery terminals and cables shall be arranged to prevent incorrect installation. Battery terminals shall be located for access in less than 30 seconds with jumper cables. Battery cables shall be flexible and sufficiently long to reach the batteries in the extended tray position without stretching or pulling on any connection. Cables shall not lie on top of batteries. The battery terminals and cables shall be color-coded with red for the primary positive and black for negative.

The battery tray shall be made of stainless steel, or polyethylene, and shall pull out easily and properly support the batteries during service, filling by manual or automatic equipment, inspection of water levels, and replacement. In the normal position, the battery tray shall not be supported by rollers. A positive lock shall retain the battery tray in the normal position. Batteries shall be easily accessible for inspection, serviceable only from outside the bus, securely mounted on a sliding or swing out stainless steel rollers, treated to prevent corrosion and be readily accessible for servicing, under the bus floor and separate from the engine compartment.

3.46.11 Battery Disconnect Switch and Jump Starting

A battery disconnect switch shall be furnished and located at an easily accessible location in the engine compartment or the battery box. Jump start positive and ground studs shall be furnished and located near the battery disconnect switch. Both the disconnect switch and jump start terminals shall be clearly labeled.

3.46.12 Modular Design

Design of the electrical system shall be modular so that each major component, apparatus panel, or wiring bundle is easily separable with standard hand tools or by means of connectors. Each module, except the main body wiring harness, shall be removable and replaceable in less than 30 minutes by a 3M mechanic. Power plant wiring shall be an independent wiring module. Replacement of the engine compartment wiring module(s) shall not require pulling wires through any bulkhead or removing any terminals from the wires.

3.46.13 Microprocessor/PLC Based Systems

The electrical system shall be controlled by a multiplex programmable logic controller that shall be located in a compartment protectedfrom moisture, dust, and dirt.
Versatility and future expansion shall be provided for by the system architecture. The multiplex wiring system shall provide and distribute power to ensure satisfactory performance of all electrical components. The system shall be capable of monitoring select bus systems including, but not limited to, door operation, engine, and transmission. The mechanical and electrical systems requiring monitoring will be determined at a post award meeting. The multiplex system shall also be in accordance with all specified modular design requirements. All electrical and electronic devices, subsystems, and components shall be repairable and maintainable by the Government.

The components of the multiplex system shall be of modular design, thereby providing for ease of replacement by maintenance personnel. The modules shall be easily accessible for troubleshooting electrical failures and performing system maintenance. Each module shall be shielded to prevent interference by EMI and RFI; and shall utilize LEDs to indicate circuit integrity and assist in rapid circuit diagnostics and verification of the load and wiring integrity. In conjunction with relays, if necessary, each circuit shall be capable of providing a current load of up to 10 Amperes. The internal controls shall be a solid state device, providing an extended service life. Wiring for data bus and node module power shall consist of three, 22 gage or larger, UL approved, shielded, twisted pairs.

Protection to each individual circuit shall be provided. A communication check system, integral to the multiplexing, shall be provided.

The system shall be hosted on a customer furnished IBM-compatible personal computer as well as a hand held field diagnostic unit, if available from the manufacturer, capable of reading the network data control function and address data, or function code.

All wire and cable insulation shall meet the flame and smoke test requirements of this specification. The contractor shall mark each wire every foot by wire type and each wire end with a function code using a scheme subject to approval by the Government.
All electrical wire and cable used in the vehicles shall be approved by the Government. Braided copper wire shall be used in all ground strap applications. Flexible stranded copper wire is acceptable in other applications. All conductors of multi-conductor cables shall be terminated. The shields on multi-conductor low voltage control cables shall be terminated on one end only to minimize the effects of ground-loops.

3.47 Materials and Workmanship

Materials for the construction of the bus vehicle shall be in accordance with a generally accepted, domestic specification or cited standard (such as ASTM, AISI, SAE, other) unless the contractor obtains written approval for a substitution from the Government. All materials shall economically and safely perform and be satisfactory within their operating environment and in accordance with their intended function. Overall work quality shall conform to the best manufacturing practices in all respects. All parts and components shall be new and in no case will used, reconditioned, repaired, or otherwise obsolete parts be accepted by the Government. Any component(s) determined to be nonconforming to the requirements herein shall not be repaired or modified for use without Government approval.
Whenever a commercial material is not covered by a recognized specification or standard, the contractor shall identify the material by the commercial trademark, name, and address of the supplier. The contractor shall submit a description of the material composition for approval by the Government.

Single-source materials shall not be permitted unless approved. Approval shall be determined on a case-by-case basis. Specification equivalency/benefit data for any substitution to a cited standard shall be submitted to the Government for review and approval.

The following materials shall not be used in the construction of the bus:

- PVC;
- Asbestos;
- Lead (in brake shoes);
- Urethane foam;
- Chlorinated fluorocarbons that may cause environmental problems or handling hazards;
- Materials that, in their normal installed state, emit products that are known to be toxic or irritative.

Bus vehicle body structural parts that are permanently covered and concealed after assembly shall not be made of copper, copper bearing aluminum alloys, brass, bronze, silver, or nickel. Foreign matter, such as shavings, chips, etc., shall be completely removed from all parts of the vehicle, its components, assemblies and subassemblies, whether hidden or exposed. Surfaces exposed to passengers, crew, or maintainers shall be smooth and free of burrs, fins, sharp edges, and dangerous protrusions.

The contractor shall keep on file material safety data sheets (MSDS) for all chemical materials (paints, solvents, adhesives, etc) used in the manufacture of the vehicles, and provide MSDS information as requested by the Government for any questionable material.

All non-metallic materials utilized in the construction of the vehicles shall be subject to the approval of the Government. The use of caulking and sealing compounds shall be minimized. Caulking and sealing compounds shall be applied in accordance with the manufacturer's instructions and recommendations, shall be non-staining, and shall be supplied in colors closely matching those of adjacent materials and surfaces. Caulking compound application shall be fully compatible with subsequent paint application (as applicable).

Threaded aluminum fasteners shall not be used, except in tapped holes in solid aluminum structures, subject to approval and review by the Government.

All name and rating plates shall be permanently attached using mechanical fasteners. Exceptions may be made for small components and circuit boards. All materials shall be new and of recent manufacture. Material which is found defective and repaired cannot be used unless specific approval is granted by the Government.

The contractor shall maintain records that trace all materials to their manufacturers, and shall help verify compliance with quality standards specified or cited in this specification. This information shall be made available to the Government upon request.

3.47.1 Units of Measure

Each vehicle system shall be designed and manufactured to a single standard. U.S. inch and ISO metric parts shall not be mixed within components.

3.47.2 Storage of Materials

All stored material subject to corrosion shall be protected by waterproof covers or coatings. Equipment covers, cable entrances, and openings shall be closed to prevent ingress of water or dirt. All dated material shall have the expiration date clearly marked. Expired material shall not be used. Rejected or damaged material shall be clearly marked and separately stored.

3.47.3 Fasteners

3.47.4 Fastening to Structural Members

The contractor shall ensure the proper application of all non-threaded and threaded fasteners. Torque marks or stripes extending from the secured hardware shall be applied to all safety related hardware, including door, and brake equipment bolts. In cases of suspect workmanship, the Government may require non-safety related hardware to have a tightening indication present which is not required to extend onto surrounding surfaces.

3.47.5 Threaded Fasteners

The number of different sizes and styles of fasteners used shall be minimized and shall meet the following requirements:

- All fasteners shall be stainless steel, dichromate, or zinc plated steel, depending on the specific application;
-
- No protruding screws, mounting bolts, or similar items shall be permitted on the exterior of the vehicle;
- On the vehicle interior, all exposed screws shall be stainless steel with flat or oval heads;
-
- Fasteners not exposed to passengers on the vehicle interior shall be of stainless steel or zinc-plated steel;
-
-

3.47.6 Locking Requirements

Locking requirements for threaded fasteners shall be in accordance with the following:

- All threaded fasteners shall be self-locking or provided with locking devices;
- Locking devices shall be either lockwire, lockwashers or locknuts as appropriate for the application or service;
- Lockwire, if used, shall be stainless steel;

- Locknuts shall be of the deformed thread prevailing torque type;
- Bolts for use with locknuts shall not be heated or drilled for cotter pins;
- All locknuts shall comply with the Industrial Fasteners Institute requirements regarding to locking ability.

3.47.7 Torque Marking/Indexing

The contractor shall ensure the proper application of all threaded fasteners. Torque marks or stripes extending from the secured hardware shall be applied to all safety-related hardware, including bus vehicle body, door, and brake equipment bolts.

3.47.8 Reserved

3.47.9 Riveting

Rivet holes shall be accurately located and aligned. Rivets shall completely fill the holes. Hand-driven steel rivets shall be driven hot and shall completely fill the holes. Mechanically driven rivets may be driven cold. Use of blind rivets shall require Government approval. All rivets shall be of a solid center type. External rivet heads shall be concentric with the body of the rivets and free from rings, pits, burrs, and fins. Surfaces exposed to passengers, operator(s), or maintenance personnel shall be smooth and free of burrs, fins, sharp edges, and dangerous protrusions.

3.47.10 Dissimilar Metal Treatment

. Isolating and moisture-proofing materials, appropriate to the materials being joined, shall be used.

Areas exposed to corrosive fluids or cleaning solutions shall be protected with coatings resistant to those fluids. The contractor shall be responsible for verifying that all such areas are protected through communications with the Government.

Except as otherwise indicated, all aluminum exposed to view in finished work in the interior of the vehicle shall have a protective anodized coating. Surfaces of aluminum alloy parts secured to ferrous parts shall be protected with one-part polysulfide or silicone sealant used as joint compound, or with joint material that is non-hygroscopic and is free from chlorides and heavy metal ions.

3.47.11 Welding

All welding practices shall be in accordance with the applicable requirements and recommendations of the American Welding Society (AWS), as contained in the latest revisions of the "Structural Welding Code" (AWS D1.1), "Aluminum Welding Code" (AWS D1.2), "Structural Welding Code - Sheet Steel" (AWS D1.3), "Recommended Practices for Resistance Welding" (AWS C1.1), and the AWS "Welding Handbook" (AWS WHB). Where non-AWS welding practices or processes will be used, such as Canadian Standards Association (CSA), the contractor shall demonstrate equivalence for Government review and approval.

The contractor shall be responsible for the quality of all welding and brazing, including the welding and brazing of its suppliers and sub-contractor's. Prior to welding, all surfaces shall be thoroughly cleaned to remove corrosion, rust, scale, slag, grease, oil, water, paint, and other foreign materials. Parts to be joined by welding shall be supported and held in position by tables, jigs, or fixtures to prevent warping.

All weld quality shall be in accord with acceptable weld criteria as defined in AWS or CSA welding codes.

The contractor shall have an on-going quality control program of inspection, non-destructive and destructive testing to insure quality welds. The contractor shall provide for Government review and approval complete documentation as to how its welders are certified and monitored. Complete documentation shall further be provided describing, in detail, the type of weld testing performed, frequency of the tests, and actions taken if defective workmanship is discovered. Documentation shall also be provided on the testing and monitoring of the welding devices utilized.

3.48 Air and Hydraulic Hoses

Air or hydraulic hose applications shall not be permitted in locations where adequate visual inspections cannot be made. Air hose installations shall be located/arranged in such a manner as to prevent accidental cross connections to other hoses located in the same general area. Hose installations shall be such that kinking, rubbing, straining, and unnecessary swinging are precluded. Routing that requires other piping or cables as the sole means of support shall not be accepted.

3.49 Air Pressure Vessels

Brake air reservoirs shall comply with SAE J10 Automotive and Off-Highway Air Brake Reservoir Performance and Identification Requirements.

3.50 Bearings and Lubrication

All bearings and lubricants shall be readily available in the United States. Standard grease fittings or plugs shall be provided for all bearings not internally splash- or bath-lubricated. All rotary shafts shall be supported by cylindrical or tapered roller bearings where practicable. Ball bearings may be used, subject to approval.

Bearings subject to atmospheric or liquid contamination shall be sealed by labyrinth, lip, or face seals. Bearings that are not splash or bath lubricated shall be provided with standard grease fittings and drain plugs or pressure-release devices for relubrication. Ball-bearings of 25 mm [1-in] shaft size and smaller may be factory lubricated-for-life, subject to approval. Bearings shall be installed and removed without major disassembly of related components, and in no case shall thrust be carried across the rolling elements in pressing a bearing into its seat.

Sleeve bearings shall be used for shafts with rotary motion of less than one full revolution. Sleeve bearings shall be adequately lubricated. Sleeve bearings supporting ferrous shafts shall be composed of bronze, brass, or aluminum alloys as approved. Sleeve bearings may be used to support rotary shafts if space limitations preclude the use of anti-friction bearings.

Self-lubricated bushings (sintered metal) shall be used in accordance with the manufacturer's recommendations, but shall not be used for shafts with speeds greater than 500 rpm.

All lubricants shall be products approved by the supplier of the parts on which the lubricant is to be used. All lubricants shall, as a minimum, conform to applicable ANSI and ASTM specifications. Multi-purpose lubricants shall be used where possible.

3.51 Flammability and Smoke Emissions

Materials used in the vehicle shall comply with the flammability, smoke emission, toxic gas, and fire retardation requirements specified herein. Those materials and products generally recognized to have high toxic products of combustion shall not be used. Materials used in the vehicles shall be zero halogen in addition to meeting the low-smoke requirements specified below. Unless specified below, all materials and construction shall meet applicable NFPA requirements. All materials used in the construction of the passenger compartment of the bus vehicle shall meet the requirements of FMVSS 302. In addition, with the exception (only) of the following list of materials, all materials used in the construction of the passenger compartment shall be tested in accordance with and meet the performance criteria in the recommended Fire Safety Practices as defined in FTA Docket 90-A, dated October 20, 1993.

Only the materials listed below are not required to meet FTA Docket 90-A:

- Adhesives, sealants and undercoating, including RTV and PPG undercoating
- Front and rear fiberglass caps
- HVAC air return grill-ABS material
- Polycarbonate sheets for driver' barrier and door modesty panels
- Driver's dash box and side console-ABS material.
- Window and door rubber seals
- Flooring material adhesive to plywood floor
- Engine compartment and under floor noise insulation
- Passenger compartment fluorescent light covers
- Driver seat components (depending on specifications)
- Passenger seat handles if not stainless steel

Materials proposed for the construction of the bus which are listed in the aforementioned FTA Docket 90 must be test certified according to the defined procedures and performance criteria in FTA Docket 90-A.

The Government will review and approve contractor submitted material certification(s) proving that the materials to be utilized in the construction of the proposed bus meets the aforementioned FMVSS and/or FTA Docket 90 requirements. All testing must be conducted by a recognized, independent laboratory testifying to material conformance. The contractor shall provide to the Government complete test report results prior to shipment of the first production bus vehicle.

3.52 General Workmanship

A**.** Buses shall be free from defects that may impair their serviceability or detract from appearance.
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B. All components will be new. Defective components shall not be furnished.
Parts, equipment, and assemblies which have been repaired or modified to overcome deficiencies shall not be furnished without the approval of the Government. Component parts and units shall be manufactured to definite standard dimensions with proper fits, clearances and uniformity. General appearance of the vehicle shall not show any evidence of poor workmanship.

C. The following shall be reason for rejection:

1. Rough, sharp or unfinished edges, burrs, seams, corners, and joints.

2. Non-uniform panels and edges that are not radiused, beveled, etc.

3. Paint runs, sags, orange peel, "fish eyes," etc., and any other imperfection or lack of complete coverage of paints or coatings.

4. Body panels or components that are uneven, unsealed, or contain cracks, dents or have voids.

5. Misalignment of body fasteners, glass, viewing panels, light housings, other items with large or uneven gaps, spacing etc. such as door, body panels and hinged panels.

6. Improperly fabricated and routed wiring or harnesses, and electrical connections.

7. Improperly supported or secured hoses, wiring harnesses, mechanical controls etc., including interference with other components.

8. Interference of chassis components, body parts, doors, etc.

9. Leaks of any gas, or fluid lines (air conditioning, coolant, oil, etc.).
10. Noise, panel vibrations etc.

11. Inappropriate or incorrect use of hardware, fasteners, components, or methods of construction.

12. Incomplete or improper welding, riveting or bolting.

13. Lack of uniformity and symmetry where applicable.

14. Loose, vibrating, abrading body parts, components, subassemblies, hoses, wiring harnesses or trim.

15. Improper body design or interface with the chassis substructure that could cause injury during normal use or maintenance, and which fail to provide access to perform routine or mandatory repairs or maintenance on vehicle electrical and mechanical systems. In addition, the improper combination of options which by their combination and installation are inherently incompatible with regard to function or safety.

16. Sagging non-form fitting upholstery or padding, holes, tears, discoloration, etc.

17. Visual deformities and equipment malfunctions.

18. Unsealed appurtenances or other body components, gaskets, etc.

19. In addition, any deviation from specification requirements or any other item, whether or not stipulated herein, that affects form, fit, function, finish, durability, reliability, safety, performance or appearance shall be cause for rejection.

3.53 Manuals and Catalogs

Operator’s instruction manuals shall be provided to the Government in the quantities as follows:

- Operator's Instructional Manual – 2 copies per bus

The Operator's Instruction Manual shall contain all information needed for the optimum operation of the bus vehicles; including general vehicle familiarization material, location, function and operation of all controls, indicators, switches, and emergency procedures and trouble diagnoses.

3.54 Warranty Requirements

Warranties in this document are in addition to any statutory remedies or warranties imposed on the Offeror. Consistent with this requirement, the Offeror shall warrant and guarantee to the Government each complete bus vehicle and specific subsystems and components according to the following provisions. All component warranties shall be manufacturer’s warranties furnished by the manufacturer of the component.

The Offeror shall, at its own expense, have one or more competent representative(s) available on request to assist the Government staff in the solution of engineering or design problems that are within the scope of the Specifications and that may arise during the warranty period.
Complete Bus Vehicle:

The complete bus vehicles shall be warranted and guaranteed to be free from defects or related defects for 2 years or 100,000 miles, whichever comes first, beginning on the date of official acceptance by the Government of each bus vehicle. During this warranty period, the bus vehicles shall maintain their structural and functional integrity. The warranty shall be based on regular operation of the bus vehicles under the operating conditions prevailing in Government service.

Chassis Structure:

The chassis structure is composed of all components that are welded together to form the main frame and body structure. Bolted-on components, unsprung suspension components, and operating hardware are considered add-ons and therefore are not a part of the basic body structure.

The chassis structure, including structural elements of the suspension attached to the body structure but excluding suspension walking beams, shall be warranted against corrosion failure and/or fatigue failure for a period of 12 years or 500,000 miles, whichever comes first.

Subsystem and Components:

Specific subsystems and components shall be warranted and guaranteed to be free from defects and related defects for the times or mileages given in Table 0-1 below, beginning on the date of official acceptance of each bus or piece of equipment. Extended warranties shall be purchased by the contractor if required to meet the required warranty years/mileage in Table 0-1. The contractor shall furnish the manufacturer’s base or standard warranty certificates for all warrantable items, and proof of purchase from the manufacturer of extended warranties, all to be included with each bus vehicle. Copies of base or standard warranty certificates and proof of purchase of extended warranties shall be furnished to the government inspector prior to acceptance of the bus by the inspector and sent to the contracting officer within 10 days after acceptance of the bus. The proof of purchase of extended warranties shall consist of at least one of the following:

- Extended warranty certificate with specific item serial number
- Invoice for extended warranty referencing specific serial number and proof of payment

Table 0-1: Subsystem and Component Warranty

		GSA Baseline		**Offeror’s Warranty**	
Item	**Description**	**Years**	**Mileage**	**Years**	**Mileage**
1	Diesel Engine (Requires purchase of extended warranty)	5	300,000		
2	Transmission	2	Unlimited		
3	Drive and non-drive axles	3	150,000		

4	Suspension	1	50,000		
5	Basic body structural integrity	3	150,000		
6	Corrosion protection	7	350,000		
7	Radiator and Charge Air Cooler	2	100,000		
8	HVAC	3	Unlimite d		
9	Destination Signs	3	150,000		
10	Door systems – Parts Only	3	150,000		
11	Wheelchair ramp	3	150,000		
12	Alternator	2	100,000		

Voiding of Warranty:

The warranty shall not apply to any part or component of the bus vehicles that has failed as a direct result of misuse, negligence, accident, or that has been repaired or altered in any way so as to affect adversely its performance or reliability, except insofar as such repairs were in accordance with the Offeror's maintenance manuals and the workmanship was in accordance with recognized standards of the industry.

The warranty on any part or component of the bus vehicles shall also be void if the Government fails to conduct normal inspections and scheduled preventive maintenance procedures on the same part or component substantially as recommended in the Offeror's maintenance manuals and such failure by the Government is the sole cause of the part or component failure.

Exceptions to Warranty:

The warranty shall not apply to scheduled maintenance items and items furnished by the Government, except insofar as such equipment may be damaged by the failure of a part or component for which the Offeror is responsible.

Detection of Defects:

If the Government detects a defect within the warranty periods, it shall promptly notify the Offeror's representative. Within 5 working days after receipt of notification, the Offeror's representative shall either agree that the defect is in fact covered by warranty, or reserve judgment until the subsystem or component is inspected by the Offeror's representative or is removed and examined at Government property or at the Offeror's plant. At that time the status of warranty coverage on the subsystem or component shall be mutually resolved between the Government and the Offeror. Work necessary to effect the repairs shall commence within 10 working days after receipt of notification by the Offeror.

Scope of Warranty Repairs:

When warranty repairs are required, the Government and the Offeror's representative shall agree within 5 working days after notification of a detected defect on the most appropriate course for the repairs and the exact scope of the repairs to be performed under the warranty. If no agreement is obtained within the 5-day period, the Government reserves the right to commence the repairs.

Repair Parts and Service

Continuous operation of the buses is of utmost importance. Parts such as body panels, window assemblies excluding glass panels and windshields, doors, seats, seat covers, engines, and transmissions shall be shipped within 30 days from order. Parts shall be available for a period not less than 10 years from the date of delivery.

3.55 Testing and Quality Assurance

3.55.1 Water Spray Test

- The OEM’s standard water spray test shall be used, subject to Government approval.
- Testing shall be conducted in accordance with the bus manufacturer’s standard testing procedures.
- Evidence of water leakage into passenger compartment shall be cause for rejection until leaks are corrected. Water intrusion through louvered rear caps into the engine compartment and rear A/C compartment is anticipated and not a cause for rejection. Drain lines shall be furnished in these compartments to allow water to drain from compartment floors and drip pans

3.56 Verification Requirements

The contractor shall verify compliance to the requirements herein prior to first bus production release. Verification may be in the form of contractor production line bill of material, drawings, instructions, test results, vendor documentation, certifications, or design analytical data. The contractor shall not proceed with production before receiving agreement from the Government that verification of compliance is satisfactory. Acceptance of designs by the Government does not relieve the contractor from the contractor's responsibility to meet all the requirements and specifications herein, whether reviewed by the Government or not. It shall be the contractor's responsibility to assure all production documentation results in the production of vehicles that are in compliance with these requirements and specifications.

PWO Provide with offer

FVI First Vehicle Inspection

The type of verification to be submitted for acceptance is listed below:

ET Engineering Test- to be performed by the contractor and results accepted by the Government. Engineering tests are tests resulting in data output from engineering instruments used to take measurements of parameters such as temperatures, sound levels, etc.

QAPVT Quality Assurance Performance and Verification Testing- to be performed by the contractor and verified by the contractor to the Government. QAPVT tests are functional tests that do not involve instrumentation. The results are observable and recorded as pass/fail, i.e. wheelchair lift platform functions as intended; lights function as intended, etc. Systems or components that fail shall be annotated with the corrective action taken and retested until passed.

The contractor shall, within 45 days after receipt of order, submit detailed plans for the performance of all ET tests. These plans, at a minimum, shall include all instruments to be used, their calibration certification, and the procedures to be followed.

The Government may accept designs at either the PWO or FVI. These acceptances shall constitute design freezes and enable the contractor to proceed with production activities relative to the accepted designs. The design freeze shall be effective for the duration of the production of the entire quantity of buses being procured.

3.57 GSA Form 1398

Contractors shall affix one copy of GSA Form 1398, GSA Purchased Vehicle, fully completed, to the right or left front door lock face or door post after the pre-delivery inspection is made at the manufacturer’s plant. All marks on windows and other labels (except the Monroney label and labels cautioning against drained transmission, crankcase, and rear axle) shall be removed. Copies of GSA Forms 1398 are available from the Contracting Officer.

NOTE TO OFFEROR: Data shown in the solicitation schedule of items may be utilized to satisfy the requirements for the following to be shown on GSA Form 1398:

(1) Receiving Agency—Examples are: State, USIA, Forest Service, and GSA.

(2) Purchase Order No.—Use the five-digit "case" or "file" number (often referred to as the RPN number)

4.0 Reference Information

4.1 Abbreviations and Terminology

The following is a list of abbreviations used in this Specification:

- ADA — Americans with Disabilities Act
- ANSI — American National Standards Institute
- ASHRAE — American Society of Heating, Refrigerating and Air Conditioning Engineers
- ASE — Automotive Service Excellence
- ASTM — American Society for Testing and Materials
- FTA — Federal Transit Administration
- FMCSR — Federal Motor Carrier Safety Regulations

- FMVSS Federal Motor Vehicle Safety Standards
- JIC Joint Industrial Council
- Government General Service Administration
- OSHA Occupational Safety and Hazard Administration
- SAE Society of Automotive Engineers
- UL Underwriter's Laboratory
- USEPA United States Environmental Protection Agency

4.2 Definitions

The following are definitions of special terms used in this Specification:

- Approach Angle - The approach angle is the angle measured between a line tangent to the front tire static loaded radius arc and the initial point of structural interference forward of the front tire to the ground.

- Breakover Angle - The breakover angle is the angle measured between two lines tangent to the front and rear tire static loaded radius and intersecting at a point on the underside of the vehicle that defines the largest ramp over which the vehicle can roll.

- Bus Body and Chassis OEM (Bus body and chassis Original Equipment Manufacturer)- The manufacturer responsible for all of the following:

1. The design and manufacture of the bus body and chassis, including, but not limited to, the bus undercarriage and bus body;
2. The final assembly of undercarriage, body, hybrid drive, and components into a complete and functional bus;

3. The warranty of all components, systems, and subsystems (excluding tires);

4. The assigning of the DoT VIN (Vehicle Identification Number).

- Class of Failures - Classes of failures are described below.

- Class 1: Physical Safety - A failure that could lead directly to passenger or operator injury or represents a severe crash situation.

- Class 2: Road Call - A failure resulting in an en route interruption of service. Service is discontinued until the bus is replaced or repaired at the point of failure.

- Class 3: Bus Change - A failure that requires removal of the bus from service during its assignments. The bus is operable to a rendezvous point with a replacement bus.

- Class 4: Bad Order - A failure that does not require removal of the bus from service during its assignments but does degrade bus operation. The failure shall be reported by operating personnel.

- Contractor - Person or persons, firm, partnership, corporation, or combination thereof that has entered into the procurement contract with the Government.
- Curb Weight - Weight of vehicle, including maximum fuel, oil and coolant; and all equipment required for operation and required by this specification, but without passengers or operator.

- dBA - Decibels with reference to 0.0002 microbar as measured on the "A" scale.

- Departure Angle - The departure angle is the angle measured between a line tangent to the rear tire static loaded radius arc and the initial point of structural interference rearward of the rear tire to the ground.

- Fireproof - Materials that will not burn or melt at temperatures less than 2,000 degrees Fahrenheit.

- Fire-Resistant - Materials that have a flame spread index less than 150 as measured in a radiant panel flame test per ASTM-E 162-90.

- Free Floor Space - Floor area available to standees, excluding the area under seats, area occupied by feet of seated passengers, areas forward or to the side of standee lines, and any floor space indicated by manufacturer as non-standee areas such as, the floor space “swept” by passenger doors during operation. Floor area of 1.5 square feet shall be allocated for the feet of each seated passenger that protrudes into the standee area.

- Gross Load – Load based on the number of seated and standee passengers, plus driver, at 150 lbs/passenger, the bus can carry without exceeding the front or rear GAWR and GVWR.

- GVW (Gross Vehicle Weight) - Curb weight plus gross load.

- GVWR (Gross Vehicle Weight Rating) - The maximum total weight as determined by the vehicle manufacturer, at which the vehicle can be safely and reliably operated for its intended purpose.

- GAWR (Gross Axle Weight Rating) - The maximum weight, as determined by the vehicle manufacturer, that can be carried on a particular axle, or set of axles, with a given rim and tire.

- Human Dimensions - The human dimensions used in this Specification are defined in SAE Recommended Practice J833.

- Industry/Service Proven – Component/System having two years of proven revenue service.

- Low Floor Bus - A bus which, between at least the front (entrance) and rear (exit) doors, has a floor sufficiently low and level so as to remove the need for steps in the aisle between the doors and in the vicinity of these doors.

- Maintenance Personnel Skill Levels - Defined below are ASE maintenance personnel skill levels used in this Specification.

    - 5M: Specialist Mechanic or Class A Mechanic Leader
    - 4M: Journeyman or Class A Mechanic
    - 3M: Service Mechanic or Class B Servicer
    - 2M: Mechanic Helper or Bus Servicer
    - 1M: Cleaner, Fueler, Oiler, Hostler, or Shifter

Note: Whenever a specific time is indicated to access components or complete a task, it is assumed the vehicle is in the location where the work is to be performed. All necessary equipment is in its correct position (tools, jacks, vehicle lifts, lighting, fluid recovery systems, etc.) and ready for use.

- Operator - Individual on board who is in charge of bus operation.

• Operator's Eye Range - The 95th-percentile ellipse defined in SAE Recommended Practice J941, except that the height of the ellipse shall be determined from the seat at its reference height.

• Seated/Standee Load – Two hundred fifteen pounds (150 lbs) for every designed passenger seating position and for the operator.

• Standee Line - A line marked across the bus aisle to designate the forward area that passengers may not occupy when the bus is moving.

• SLW (Seated Load Weight) - Curb weight plus seated load.

• Standards - Standards referenced in this Specification are the latest revisions unless otherwise stated.

• Structure - The structure shall be defined as consisting of all the components that are welded together to form the main frame (skeleton) and body construction including axle and suspension attachment points.

• Wheelchair - A mobility aid belonging to any class of three or four-wheeled devices, usable indoors, designed for and used by individuals with mobility impairments, whether operated manually or powered. A "common wheelchair" is such a device that does not exceed 30 inches in width and 48 inches in length measured two inches above the ground, and does not weigh more than 600 pounds when occupied.

4.3 First Vehicle Inspection (FVI) and Pre-Inspection Test Drive

The contractor shall furnish a first production vehicle for inspection by Government representatives within the time frame specified by the Contract. Production methods shall assure that subsequent vehicles are identical to the approved first production vehicle. The first production vehicle shall be complete in every respect, i.e., all components, equipment and accessories assembled and/or installed.

The contractor shall inspect the bus for compliance to all the requirements herein, including all workmanship requirements, prior to Government inspection. The purpose of the Government inspection is primarily to verify the contractor's inspection and corrective actions. Prior to the Government inspection, the contractor shall furnish the Government inspector verification that the inspection has been made. This verification shall, at a minimum, consist of the following:

- In –process and final inspection documentation
- Non-compliances found during in-process and final inspection

- Disposition of all non-compliances and rejections (corrective actions)
- Test results as required herein
- Performance Assurance and Verification results as required herein
- Test drive results, inspection, and corrective actions
- Deficiencies discovered by the Government inspector on the vehicle presented for first vehicle inspection may subject the contractor to reinspection charges.

It shall be the responsibility of the contractor to verify to the Government Inspector that all tests and inspections have been made and that the vehicle presented meets all the design, performance, material, and workmanship requirements herein.

After approval of the first production vehicle, no substitution of materials, components, or assemblies will be permitted except by modification to the contract. Prior to the first vehicle inspection, the bus shall be driven on a road test by the contractor for a minimum of 20 miles. A portion of this test may be simulated on a chassis dynamometer. The road test shall be comprised of the following:

- Prior to the road test, diagnostics shall be obtained and printed from the engine ECM. Any existing fault codes shall be noted, diagnosed, corrected, and cleared.
- 20 miles of continuous state or interstate highway driving at speeds of 50-65 MPH. The bus shall be stopped a minimum of 5 times at highway turnarounds and/or other appropriate and safe places.
    - Each stop shall include one door opening/closing cycle of both doors.
    - The wheelchair ramp shall be activated at each stop.
- At the end of the test drive, another diagnostic reading from the engine ECM shall be printed. Any fault codes generated from the test drive shall be noted, diagnosed, the cause(s) corrected, and cleared.
- The bus shall be inspected after the test drive for the following:
    - Fluid leaks- (defined as fluid seepage resulting in a drop being formed) shall be recorded and corrected.
    - Pneumatic, hydraulic, HVAC, or coolant hose abrasion against bus structure or components. Hose routing deficiencies from this inspection shall be recorded and corrected.
    - Components that loosened, rattled, or otherwise caused reason for notation during the test drive. Corrective action shall be taken and recorded.

The contractor shall document all incidents, malfunctions, and any unusual bus performance noted during this test drive, plus the quantity of any fluids added during the drive. All other corrective actions resulting from the 20-mile test drive shall be noted. Prior to acceptance by the Government, all fault codes shall be corrected and cleared from the ECM.

4.4 Production Vehicle Final Inspection and Test Drive

Each production bus shall undergo a final inspection by a Government Inspector.
Prior to inspection and acceptance by the Government, each production vehicle shall be road test driven for a minimum of 20 miles as required in section 4.3 above. All contractor inspection and verification of contractor inspection requirements in section 4.3 above shall apply to each production bus. Deficiencies discovered by the Government inspector on any vehicle presented for final inspection may subject the contractor to reinspection charges.

4.5 Option Code Definitions and Requirements

The following are descriptions of possible selected option codes.

4.5.1 Option HEVA Allison E^p 40 two mode, compound split parallel hybrid drive system

If option HEVA is selected, the contractor shall furnish the Allison E^p 40 two mode, compound split parallel hybrid drive system.

4.5.2 Option BR2 Two Bicycle Front Rack

If option BR2 is selected, the contractor shall furnish and install a Sportsworks model Velo-Porter 2, or equal, demountable rack for transporting two bicycles on the front of each bus vehicle. The mounted bike rack shall not interfere with attaching towing equipment or accessing towing connectors and it shall not obstruct the driver's field of vision.

4.5.3 Option BR3 Three Bicycle Front Rack

If option BR3 is selected, the contractor shall furnish and install a Sportsworks model Apex 3, or equal, demountable rack for transporting three bicycles on the front of each bus vehicle. The mounted bike rack shall not interfere with attaching towing equipment or accessing towing connectors and it shall not obstruct the driver's field of vision.

4.5.4 Option SRCB Stop Request Cables

If option SRCB is selected, a passenger "Stop Requested" signal system that complies with applicable ADA requirements defined in 49 CFR, Part 38.37 shall be provided. The system shall consist of a heavy-duty pull cable, chime, and interior sign message. The pull cable shall be located the full length of the bus on the sidewalls at the level where the transom is located. If no transom window is required, height of pull cable shall approximate this transom level and shall be no greater than 63 inches as measured from floor surface. It shall be easily accessible to all passengers, seated or standing. At each wheelchair parking position and priority seating positions additional provisions shall be included to allow a passenger in a mobility aid to easily activate “Stop Requested” signal

4.5.5 Option FBOX Fare Box

If option FBOX is selected, a fare box shall be provided. The offeror shall identify the makes/models/features that the ordering activity may select from.

4.5.6 Option FFPS Forward Facing Passenger Seating

If option FFPS is selected, the contractor shall provide forward facing passenger seating equipped without seatbelts. Note: GSA Option FBOX (fare box) is required with FFPS to comply with DOT-NHTSA-FMVSS.

The seats shall be Freedman CitiPro, American Seating Insight, or equal, with stainless steel frames with padded inserts (meeting FMVSS 302 and FTA Docket 90-A). The successful offeror shall furnish a seating layout drawing which will be approved by the Government.

Seat upholstery design and color shall be a stain hiding blend and shall be the OEM’s standard upholstery complimenting the interior color scheme.

As applicable, hip-to nee room measured from the front of one seat back horizontally across the highest part of the seat to the seat or panel immediately in front shall be no less than 26 inches. At all seating positions in paired transverse seats immediately behind other seating positions, hip-to-knee room shall be no less than 26 inches. Seat height, except for rear bench seat, shall be 17 inches. The rear bench seating height shall not exceed 21 inches.

Transverse mounted seats shall, wherever possible, be fully cantilever mounted.

4.5.7 Option PSM Parts and Service Manuals

If option PSM is selected, the following shall be furnished as one set for the entire order (not per bus):

- Preventive Maintenance Manual – two copies, one with laminated pages.
- Parts Catalog – two copies, one with laminated pages.
- Service Manual-two copies, one with laminated pages, consisting of the following:
    - Electrical Schematics Manual
    - System Interface Diagrams-Electrical
    - Integrated Wiring Diagrams

- Major sub-contractor manuals (engine, transmission, wheelchair lift, HVAC)
- All publications submitted as hard copies shall, wherever possible, also be submitted in electronic format in the form of CD-ROM disks.

Preventive Maintenance Manual

The Preventive Maintenance Manual shall enable the maintainer to perform the periodic inspection and preventive maintenance tasks. These shall include all lubrication and inspection requirements for all apparatus requiring such work on a periodic basis to maintain the bus vehicles in satisfactory working order.
The Diagnostic and Troubleshooting section of the manual shall enable the maintainer to troubleshoot, adjust and complete running repair for all systems on the bus vehicles. Running repair is the diagnosis and correction of any subsystem or component malfunction by adjustment, repair or replacement in order to return the vehicle to service in a reasonably short period of time. The section shall include a general description and operation of each system, block diagrams, signal flow diagrams, detailed schematics with narrative and functional wiring and piping diagrams.

The manual shall contain a detailed description of components of the bus to enable a maintainer to service, maintain, and repair the vehicle. The manual shall include a complete systematic procedure for troubleshooting and long term periodic maintenance requirements for all components.

Parts Manual

The Parts Manual shall enumerate and describe components with its related parts and contractor part numbers. Cut-away and exploded drawings shall be used to permit identification of all parts not readily identified by description. Each part or component shall be identified as being part of a functional group.

An important aspect of the parts catalog shall be the complete itemization of all servicing materials (oils, paints, special compounds, greases, other) required on the bus vehicles and the component requiring its use.

Sub-contractor/supplier manuals shall be purchased and may be furnished “as is”. Within the manuals, the bus vehicles shall be treated as a whole and not as a grouping of disassociated parts from various suppliers. It shall be the responsibility of the bus OEM to insure that all of the suppliers' subsystems are presented in sufficient detail to present a complete and clear picture of the whole vehicle and that terms and functional designations of wires and components are consistent throughout. The material in all manuals shall be identically organized and indexed with compatible numbering.

4.5.8 Option DIAG Diagnostic Software and Hardware

If option DIAG is selected, the following shall be furnished for the order (not per bus):

- One set of multiplex electrical system service and diagnostic software
- One hand held multiplex diagnostic tool, if available from the multiplex system manufacturer.

4.5.9 Option OTRN Operator Training

If option OTRN is selected, the contractor shall provide a minimum of 16 hours of operator training necessary for the ordering agency to safely and reliably operate the bus. Training sessions shall include training on all bus systems, subsystems, and included optional systems.

4.5.10 Option MTRN Maintenance Training

If option MTRN is selected, the contractor shall provide aq minimum of 40 hours of maintenance training to the ordering agency's maintenance personnel covering, at a minimum, engine, transmission, electrical, HVAC, fuel system, and contractor proprietary system maintenance training.

4.5.11 Option MIL Military Markings and Data Plates

When code MIL is specified and available under contract, the following shall be furnished:
1. Data Plate. A data plate only shall be furnished. No other exterior military markings shall be furnished. At a minimum, the Data Plate shall be as follows:

NOMENCLATURE ____
MAKE AND MODEL ____
CHASSIS SERIAL NUMBER (VIN) _________________________________
VEHICLE CURB WEIGHT: _________________________________LBS
PAYLOAD, MAXIMUM: __________________________________LBS
GROSS VEHICLE WEIGHT RATING, MAX.____________________LBS
GROSS COMBINATION WEIGHT RATING, MAX.________________LBS
DATE OF DELIVERY: _______________MONTH_______________YEAR
WARRANTY: ___________MO. ___________________________MILES
CONTRACT NUMBER ____
RPN NUMBER ____

2. A DD Form 250 (MATERIAL INSPECTION AND RECEIVING REPORT) shall be furnished in lieu of a GSA form 308.

4.5.12 Option PSMA Parts and Service Manuals, 2 sets, Ship to WR ALC/LEET

When code PSMA is specified, two sets of parts & service manuals (one paper and one electronic) shall be sent to Warner Robins Air Force Base as detailed in the vehicle vendor’s contract. Each set of manuals shall contain a DD250.

4.5.13 Option RSLV Roof Skylight Ventilators With Safety Glass Lid

When option RSLV is specified, the contractor shall furnish the following:

A quantity of two or three roof vents, Spheros Glass Hatch Vision modular roof hatch, or equal, with transparent single pane UV blocking/anti-glare safety glass lid compliant with FMVSS 205 or 2001/85EC and EC94/28, installed approximately equally spaced along the longitudinal centerline of the bus, with the forward vent installed approximately over the front axle. The offeror shall propose the exact quantity and spacing along the bus centerline compatible with HVAC capacity, roof construction, and roof cross bow locations, subject to final approval by the Government. Orientation of these vents may be either with the longer dimension parallel or perpendicular to the longitudinal centerline of the bus. The purpose of these vents is to provide skylights to maximize the viewing experience of park visitors and to increase bus fresh air ventilation capability. Escape hatch requirements are defined in paragraph 3.43.6 above.

4.5.14 Option PSUA- Passenger Seating Upgrade to American model

When code PSUA is specified, passenger seating shall be upgraded to American model Vision, 34 seating configuration with barrier.

4.5.15 Option PSDK- Passenger Seating Downgrade to Kiel model

When code PSDK is specified, passenger seating shall be downgraded to Kiel series, 34 seating.

4.5.16 Option HIF- Interior Floor Heaters

When code HIF is specified, interior floor heaters on the street and curb sides shall be provided.

4.5.17 Option AH- Auxiliary Heater

When code AH is specified, interior auxiliary heaters shall be provided.

4.5.18 Option TSAR- Spare Wheel/Tire Assembly

When code TSAR is specified, a spare wheel/tire assembly shall be provided. The assembly shall be identical to the rear tires, shall be furnished, and shipped loose. The tire shall be wrapped in stretch shipping plastic or card board and held in position to prevent it from abrading the bus interior during delivery.

4.5.19 Option STHE- Electric Starter for Hybrid Electric

When code STHE is specified, a conventional flywheel mounted electric engine starter motor shall be provided. This availability of this option is dependent the hybrid propulsion type.

4.5.20 Option CPT – Custom Paint Color

When code is specified, custom color paint shall be provided.

4.5.21 Option UPHL – Custom Upholstery

When code UPHL is specified, custom upholstery shall be provided and installed.

4.5.22 Option NPSG National Park Service Exterior Graphics

Each bus shall be furnished with an exterior graphics package in accordance with the following requirements:

- Each bus shall be furnished with an exterior multi-color graphics system designed for long term use in extreme conditions. The graphics system shall have a minimum 5 year warranty against fading, cracking, and peeling.
- The graphics system shall be a 3M Scotchprint® Matched Component System compatible with the final graphic design(s).
- The graphics shall be furnished and installed only by 3M Scotchprint® licensed graphics suppliers and installers.
- It is anticipated that the bus graphics system will cover approximately 30% of the driver and passenger sides of the bus.

The exterior paint color and graphics design will be similar to the graphics in the illustrations/pictures below. The exact final graphics design will be decided at a post order meeting. The contractor shall create a drawing similar to the exterior graphics drawing below for approval by the government, and approval given, prior to applying the exterior graphics.

BUS-TRANSIT-HEAVY DUTY-LOW FLOOR-40 FT.-34 PASSENGER –DIESEL ELECTRIC HYBRID

BUS-TRANSIT-HEAVY DUTY-LOW FLOOR-40 FT.-34 PASSENGER –DIESEL ELECTRIC HYBRID

BUS-TRANSIT-HEAVY DUTY-LOW FLOOR-40 FT.-34 PASSENGER –DIESEL ELECTRIC HYBRID

BUS-TRANSIT-HEAVY DUTY-LOW FLOOR-40 FT.-34 PASSENGER –DIESEL ELECTRIC HYBRID

BUS-TRANSIT-HEAVY DUTY-LOW FLOOR-40 FT.-34 PASSENGER –DIESEL ELECTRIC HYBRID

Bus Exterior Graphics-Illustrations and Pictures for Reference Only
(Final exterior graphics design shall be decided at a post order meeting.)